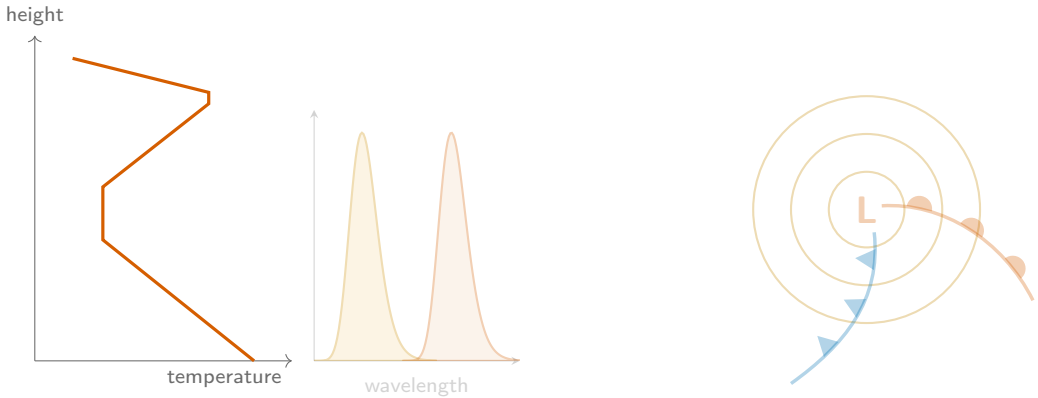
The Atmosphere: Part I

Vertical structure of the atmosphere

A worked, step-by-step guide with figures, examples and answers



Part I Part II Part III

Part I of The Atmosphere

One part of The Atmosphere, with the same numbering as the full book. Every question, data table, figure and worked answer is on the page.

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Chapter 1

How to use this book

1.1 What is in it

This book teaches the atmosphere through questions you work yourself. It has three parts, each following on from the one before: Part I describes the structure of the atmosphere, Part II explains how the Sun heats it, and Part III how that uneven heating fills it with moisture and sets it in motion. Everything you need is on the page: every question, data table, map and figure, with its worked answer.

Table 1.1. The three parts of the book.

Part	What it covers	Chapters	Appendix
I	**Vertical structure of the atmosphere.** Pressure and the halving rule, oxygen at altitude, the standard atmosphere, soundings, inversions and the tropopause, lapse rates, stability, climates.	2–10	A
II	**Solar radiation.** The laws of radiation, black bodies and planetary temperatures, the Earth's energy balance and the greenhouse effect, sun angle and the seasons.	11–19	B
III	**Moisture, motion and weather maps.** Humidity and the dew point, the forces that move the air, the geostrophic wind and friction, station models, streamlines, fronts and a full weather-map analysis.	20–30	C

Every part opens with a chapter that maps each question to the chapter that answers it, and closes with a chapter that gathers all the answers in one table, a list of mistakes to avoid and a self-test. Its appendix holds the formulas, derivations and symbols. Appendix D lists the shared constants and units, and the references are at the end.

1.2 How each chapter is laid out

1. **The ideas first,** in plain language. Every important equation is numbered, for example (3.4), and followed by a *where* list of its symbols, units and values.
2. **The question**, in a box, with its data beside it.
3. **The method**, one step per line, with every number and unit shown.
4. **The answer**, in a green box, followed by a **check** that it is sensible.
5. **Worked examples**, **key ideas**, points to **take care** over, and **questions that come up**.
6. **Try it yourself**: a practice set, with answers in the green box that follows.
7. **Python**: the equations as functions, which you can run and change. They are also collected in one tested package.

How to get the most out of the book

Cover the green answer box, work the question on paper, then uncover it. If your answer differs, find the step where you diverged before reading on. That step is where the learning is.

1.3 Conventions used throughout

- **Numbering.** "Eq. (4.2)" is equation 2 of Chapter 4; figures and tables are numbered the same way. "§ 4.3" is section 3 of Chapter 4. Appendices are lettered A to D, one per part and one for shared constants.
- **Units** are SI, written as powers: $\mathrm{W\,m^{-2}}$ means watts per square metre, $\mathrm{m\,s^{-1}}$ metres per second. Pressure is in hectopascals, which are the same as millibars ($1\ \mathrm{hPa} = 1\ \mathrm{mb} = 100\ \mathrm{Pa}$). Heights are in metres or kilometres.
- **Temperature.** Temperatures in the data are in degrees Celsius. Any formula in which temperature is multiplied, divided or raised to a power (the gas law, the radiation laws, the potential temperature) needs kelvin: $T_\mathrm{K} = T_{\circ\mathrm{C}} + 273.15$. A *difference* of temperature is the same in both scales, so lapse rates can be written in $^\circ\mathrm{C\,km^{-1}}$ or $\mathrm{K\,km^{-1}}$.
- **Constants.** Each part uses rounded values, stated where they first appear; Appendix D gives precise values and sources.
- **Rounding.** Intermediate results keep at least four significant figures; answers are rounded at the end, usually to three or four.
- **Cross-references between parts** are written in the same way as within a part, with the part named: "Part II, § 17.7".
- **References** to the literature are given as author and year, for example Stull (2017), and listed in full at the end.

1.4 What to have beside you

A calculator with powers, logarithms and trigonometric functions, set to **degrees**; graph paper, a ruler, a soft pencil and three or four coloured pencils, so that each line on a graph has its own colour. Python is optional.

1.5 Acknowledgement

The questions and data in this book grew out of the practicals of EGM305 The Atmosphere at Ulster University, and are reused with the permission of the module staff. The explanations, figures, code and any errors are the author's.

1.6 Reading

Good textbooks for going further are Lutgens et al. (2019) and Ahrens and Henson (2022). A free textbook with worked examples is Stull (2017). For more depth, Wallace and Hobbs (2006) treats the physics at the next level up.

Part I

Vertical structure of the atmosphere

Chapter 2

Part I at a glance

2.1 What this part covers

This part is about the *vertical structure of the atmosphere*: how the amount of air, its pressure, its oxygen and its temperature change as you go up. It is built around fourteen questions, which fall into four groups, each needing the one before it. Every question appears in a box where it is answered, with the data it needs beside it, then a method, a figure where one helps, and a worked answer. Try each question before reading on.

How much air is above you (Questions 1–4). A simple rule, "half the atmosphere is above you for every 5.6 km you climb", gives the fraction of the atmosphere and the pressure at any height. You draw that relationship, read values from it, and use it to find how much oxygen there is on Everest or in an aircraft cabin.

How the temperature changes (Questions 5–8). You build the standard atmosphere, plot two real weather-balloon soundings beside it, and find the inversions and the tropopause on each.

Lapse rates and the shape of the troposphere (Questions 9–11). You measure a lapse rate from data, add a tropical sounding, and explain why the troposphere is thickest over the tropics.

Why the structure holds together (Questions 12–14). You explain why pressure falls fast near the ground and slowly high up, why the cold air aloft does not sink, and why Northern Ireland is milder than Newfoundland in March.

Table 2.1. Where each question is answered, and the equations it needs.

Question	What it asks	Chapter	Equations
1, 2	percentage of the atmosphere above, and pressure, at each height; a table and a graph	3	(3.5), (3.6)
3	read five heights off the graph	4	(3.4)
4	partial pressure of oxygen at those heights	4	(4.1)
5	the standard atmosphere	5	(5.1), (5.2)
6, 7, 8	plot the soundings; inversions; tropopause	6	(5.1)
9, 10, 11	lapse rate over Hillsborough; Florida; thickness of the troposphere	7	(5.1)
12, 13	why pressure falls more slowly aloft; why cold air aloft does not sink	8	(3.2)–(8.3)
14	Hillsborough against St John's	9	–

Chapter 10 gathers every answer in one table. The formulas, derivations and symbols are in Appendix A.

Chapter 3

How much air is above you: the halving rule

This chapter answers Questions 1 and 2.

3.1 Air has weight

Air is matter: every cubic metre near the ground has a mass of about 1.2 kg. Gravity pulls all of it towards the Earth, so every layer of the atmosphere rests on the layers below. **Atmospheric pressure** at any height is the weight of all the air above, pressing on each square metre. At sea level it averages 1013.25 hPa (NOAA et al., 1976); this part rounds it to 1000 hPa.

Units of pressure

Pressure is force per unit area. The SI unit is the pascal, 1 Pa = 1 $\mathrm{N\,m^{-2}}$. Meteorologists use the **hectopascal**, 1 hPa = 100 Pa, which is exactly the same as the older **millibar**: 1 hPa = 1 mb. Sea-level pressure is about 1000 hPa = 100 000 Pa.

How heavy is the air above you? A pressure of 100 000 Pa means a force of 100 000 N on every square metre. Dividing by the acceleration due to gravity, $g = 9.81\ \mathrm{m\,s^{-2}}$, gives the mass that exerts it:

$$\frac{100\,000\ \mathrm{N\,m^{-2}}}{9.81\ \mathrm{m\,s^{-2}}} = 10\,200\ \mathrm{kg\,m^{-2}}.$$

About ten tonnes of air sit above every square metre of the ground. We do not feel it because the pressure acts equally in all directions and the air inside our bodies pushes back.

3.2 Air is squeezed by its own weight

Water is almost incompressible: the bottom of the ocean is hardly denser than the top. Air is very compressible, and the air near the ground carries the weight of all the air above it, so it is squeezed the most. The lowest layers are therefore the densest, and density (and pressure) fall with height. How fast? Two physical laws answer that.

The ideal gas law links the pressure, density and temperature of a gas:

$$p = \rho R_d T. \tag{3.1}$$

where

p	pressure (Pa)	
ρ	density of the air ($\mathrm{kg\,m^{-3}}$)	about 1.2 near the ground
R_d	gas constant for dry air ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	287, Table D.1
T	absolute temperature (K)	

At a fixed temperature, pressure and density are proportional: squeeze the air to twice the density and its pressure doubles.

Hydrostatic balance. Picture a thin horizontal slab of air, one square metre in area and Δz thick. Its weight is its mass times g, that is $\rho\,\Delta z\,g$. The slab stays where it is (it does not fall or fly upwards) because the pressure pushing up on its underside exceeds the pressure pushing down on its top by exactly its weight. So the pressure must fall by $\rho g\,\Delta z$ across the slab (Wallace and Hobbs, 2006, ch. 1):

$$\frac{\Delta p}{\Delta z} = -\rho g. \tag{3.2}$$

where

Δp	change in pressure across the slab (Pa)	negative going up
Δz	thickness of the slab (m)	
ρ	density of the air in the slab ($\mathrm{kg\,m^{-3}}$)	
g	acceleration due to gravity ($\mathrm{m\,s^{-2}}$)	9.81

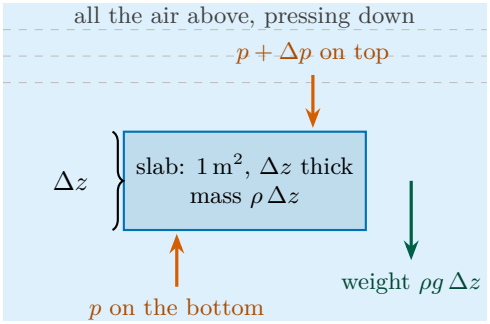


Figure 3.1. Hydrostatic balance. The slab neither rises nor falls, so the upward push of the pressure on its underside, p, exceeds the downward push on its top, $p + \Delta p$ (with Δp negative), by exactly its weight: $-\Delta p = \rho g\,\Delta z$, Eq. (3.2).

Near the ground $\rho g = 1.2 \times 9.81 = 11.8\,\mathrm{Pa}$ per metre, so the pressure falls by about 1 hPa for every 8.5 m you climb. You can feel this in your ears in a lift or on a hill.

3.3 Why the fall is exponential

Put the two laws together. By Eq. (3.2) the pressure falls by an amount proportional to the density, and by Eq. (3.1) the density is proportional to the pressure. So *the pressure falls by an amount proportional to the pressure itself*: where there is a lot of air, it falls fast; where there is little, it falls slowly. A quantity that shrinks in proportion to itself decays **exponentially**, like a radioactive substance or money losing value at a fixed rate of inflation. If the temperature is taken as constant, the result is (Appendix A.2.1)

$$p(z) = p_0\, e^{-z/H}, \qquad H = \frac{R_d T}{g}. \tag{3.3}$$

where

$p(z)$	pressure at height z (hPa)	calculated
p_0	pressure at sea level (hPa)	1000, rounded
z	height above sea level (km)	
H	scale height: the height over which the pressure falls by a factor $e = 2.718$ (km)	8.1, for the halving rule
T	a representative temperature of the air column (K)	

The **scale height** H is about 7 to 8.5 km for the temperatures of the lower atmosphere.

The halving rule is the same law

Any exponential decay has a fixed *halving distance*: the height over which the pressure halves is $z_{1/2} = H \ln 2 = 0.693\,H$. The halving rule, "for every 5.6 km you go up, there is half the atmosphere above you", is Eq. (3.3) with $z_{1/2} = 5.6$ km, that is $H = 5.6/0.693 = 8.08$ km. Written with halvings instead of powers of e, the fraction of the atmosphere above height z is

$$f(z) = \left(\tfrac{1}{2}\right)^{z/5.6} = 2^{-z/5.6}, \tag{3.4}$$

where

$f(z)$	fraction of the whole atmosphere that lies above height z (0 to 1; multiply by 100 for %)	calculated
z	height above sea level (km)	
5.6	the halving height (km)	given

and after n whole steps of 5.6 km (that is, at $z = 5.6\,n$ km) it is simply

$$f_n = \frac{1}{2^n}, \qquad \text{percentage above} = \frac{100\,\%}{2^n}. \tag{3.5}$$

3.4 Question 1: the percentage of the atmosphere above you

The question

"For every 5.6 km you go up, there is half the atmosphere above you as when you started." Keeping this rule in mind, set out a table with a row for sea level and for every 5.6 km up to 50.4 km, calculate the percentage of the atmosphere above each height, and draw the result (x-axis: percentage of the atmosphere above (%); y-axis: height above sea level (km)).

Method.

Step 1. Start at sea level with 100 % of the atmosphere above you.

Step 2. Each row of the table is 5.6 km higher than the last, so add 5.6 km to the height and halve the percentage, Eq. (3.5): $100 \to 50 \to 25 \to 12.5 \to \ldots$

Step 3. Continue until the percentage is well below 1 %: ten rows take you to 50.4 km.

Step 4. Plot each row as a point, percentage across and height up, and join the points with a *smooth* curve (Figure 3.2).

Answer to Question 1

100, 50, 25, 12.5, 6.25, 3.13, 1.56, 0.78, 0.39 and 0.20 % of the atmosphere is above sea level, 5.6, 11.2, 16.8, 22.4, 28.0, 33.6, 39.2, 44.8 and 50.4 km (Table 3.1), drawn as the curve of Figure 3.2.

Check. Half the air is below 5.6 km and three-quarters below 11.2 km, which agrees with the common statement that about half the mass of the atmosphere lies within the lowest 5 km or so. The curve never reaches zero: each step halves what is left, so there is always a little air above you. That is why there is no sharp "top" to the atmosphere.

Table 3.1. The halving rule, row by row (Questions 1 and 2). The last column is the U.S. Standard Atmosphere (NOAA et al., 1976), for comparison only.

Step n	Height (km)	% above	Pressure (hPa)	Standard atmosphere (hPa)
0	sea level	100	1000	1013
1	5.6	50	500	498
2	11.2	25	250	219
3	16.8	12.5	125	91
4	22.4	6.25	62.5	38
5	28.0	3.13	31.3	16
6	33.6	1.56	15.6	6.8
7	39.2	0.78	7.8	3.0
8	44.8	0.39	3.9	1.4
9	50.4	0.20	2.0	0.6

A question that comes up

Why do I get a curve and not a straight line?
Because equal steps up remove equal *fractions*, not equal *amounts*. The first 5.6 km removes 50 percentage points; the next removes 25; the next 12.5. Plotted against height, that is a curve that is steep near the ground and flattens higher up, the shape of every exponential decay. Plotted with a logarithmic pressure axis it would become a straight line (Figure 8.1).

3.5 Question 2: adding a pressure axis

The question

As the density of the atmosphere drops by 50 % when we go up 5.6 km, the pressure also drops by 50 %. At sea level the pressure is approximately 1000 hPa. Add the pressure for each of your heights as another column of your table and add a second x-axis to your graph to show atmospheric pressure.

Why pressure follows the percentage. Pressure is the weight of the air above (§ 3.1). If only half the atmosphere is above you, only half its weight presses down, so the pressure is half the sea-level pressure. In general

$$p(z) = p_0 \times f(z) = 1000 \text{ hPa} \times f(z). \tag{3.6}$$

where

$p(z)$	pressure at height z (hPa)	calculated
p_0	sea-level pressure (hPa)	1000, rounded
$f(z)$	fraction of the atmosphere above, Eq. (3.4)	Question 1

With $p_0 = 1000$ hPa the pressure in hectopascals is simply ten times the percentage.

The second axis. Because the pressure is exactly ten times the percentage, the pressure axis is the percentage axis relabelled: 0 to 1000 hPa along the top where 0 to 100 % runs along the bottom (Figure 3.2). The same curve then answers both questions.

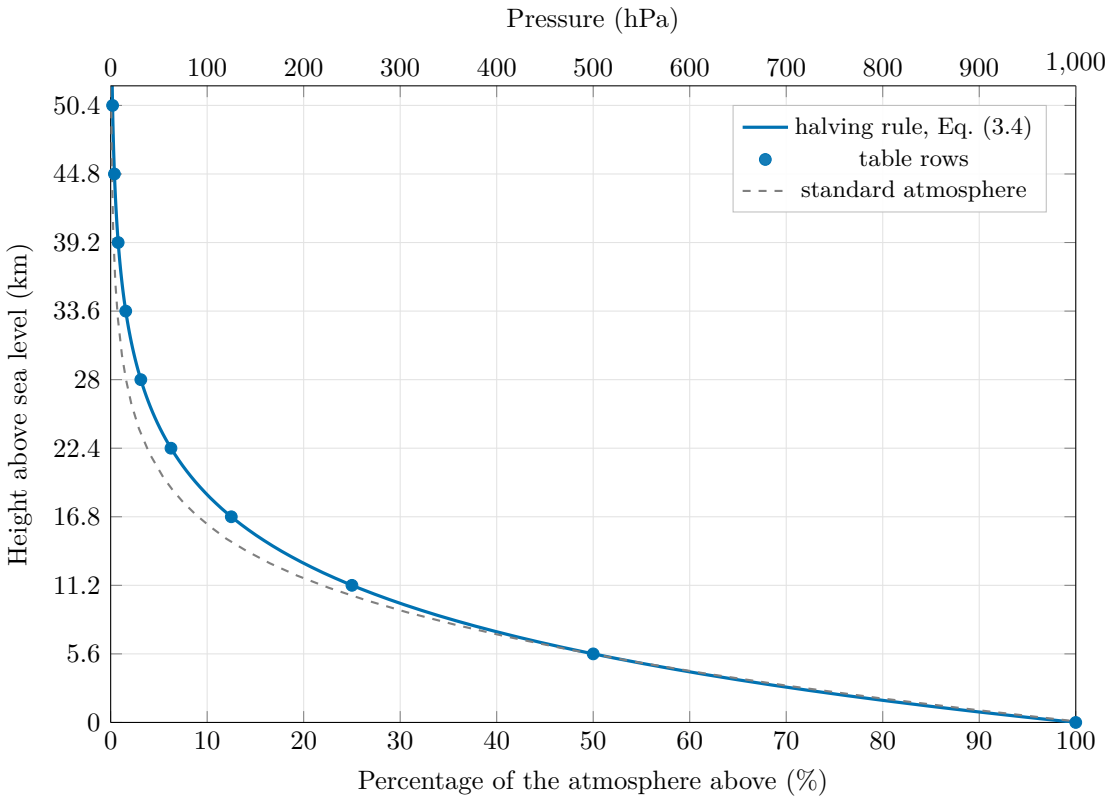


Figure 3.2. Questions 1 and 2. The percentage of the atmosphere above each height (bottom axis) and the pressure (top axis) by the halving rule. Each step of 5.6 km up halves both. The dashed line is the U.S. Standard Atmosphere: the rule is very close near the ground and overestimates the pressure higher up (§ 3.6).

Answer to Question 2

Pressure = 10× the percentage above: **1000, 500, 250, 125, 62.5, 31.3, 15.6, 7.8, 3.9 and 2.0 hPa** at sea level, 5.6, 11.2, …, 50.4 km (Table 3.1). The second axis runs from 0 to 1000 hPa above the 0 to 100 % axis.

3.6 How good is the halving rule?

The last column of Table 3.1 compares the rule with the U.S. Standard Atmosphere, an idealised but realistic average atmosphere (NOAA et al., 1976). Up to about 6 km the two agree within 1 %. Higher up the rule gives too much pressure: 250 against 219 hPa at 11.2 km, 125 against 91 hPa at 16.8 km.

The reason is the temperature in the scale height, $H = R_d T/g$, Eq. (3.3). The rule's $H = 8.08$ km corresponds to a constant temperature of

$$T = \frac{gH}{R_d} = \frac{9.81 \times 8080}{287} = 276 \text{ K } (3\ ^\circ\text{C}),$$

a reasonable average for the lowest few kilometres. But above about 8 km the air is far colder (around 217 K, −56 °C, Chapter 5), its scale height is only about 6.3 km, and the pressure falls faster than the rule assumes. Cold air is denser at the same pressure, Eq. (3.1), so a given weight of it is packed into a thinner layer. Question 12 lets you measure this from the soundings (§ 8.1).

A question that comes up

Question 2 says the *density* halves every 5.6 km. Is that the same as the pressure halving?
Only if the temperature does not change with height, because by Eq. (3.1) $\rho = p/(R_d T)$. In the real troposphere the temperature falls as you go up, so the density falls a little more slowly than the pressure (the cooler air is relatively denser). In the standard atmosphere the density at 5.6 km is 56 % of its sea-level value while the pressure is 49 %. The rule is about pressure, the weight of the air above; it is the right one to use for Questions 2 to 4.

A question that comes up

Where does the atmosphere end?
It thins out gradually and has no top. By convention the edge of space is put at 100 km, the Kármán line, where by the rule only about $2^{-100/5.6} \approx 0.0004$ % of the atmosphere remains (the real figure is smaller still). Very few gas molecules remain beyond it.

3.7 Try it yourself: Practice Set 3

Practice Set 3: pressure and height

Use the halving rule unless told otherwise; $g = 9.81\,\mathrm{m\,s^{-2}}$, $R_d = 287\,\mathrm{J\,kg^{-1}\,K^{-1}}$.

3.1 What percentage of the atmosphere is above you at 16.8 km, and what is the pressure there?

3.2 What fraction of the whole atmosphere lies between 5.6 and 11.2 km?

3.3 At what height does the pressure fall to 100 hPa?

3.4 What mass of air lies above each square metre at sea level, where the pressure is 1013 hPa?

3.5 What would the halving height be in an atmosphere at a constant 250 K?

Answers to Practice Set 3

3.1 Three halvings: 12.5 % and 125 hPa, Eqs. (3.5) and (3.6).

3.2 50 % is above 5.6 km and 25 % above 11.2 km, so $50 - 25 = 25$ % lies between them.

3.3 100 hPa is a tenth of 1000 hPa: $2^{-z/5.6} = 0.1$, so $z/5.6 = \log_2 10 = 3.32$ and $z = 18.6$ km, Eq. (3.4).

3.4 $101\,300/9.81 = 10\,330$ kg: about 10.3 tonnes.

3.5 $H = 287 \times 250/9.81 = 7.31$ km, so $z_{1/2} = 0.693 \times 7.31 = 5.07$ km. Colder air halves its pressure over a shorter distance.

3.8 The halving rule in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def fraction_above(z_km, halving_km=HALVING_KM):
    """Fraction of the atmosphere above height z_km: half for every halving_km."""
    return 2 ** (-z_km / halving_km)
```

```
def pressure_halving(z_km, p0_hpa=1000.0):
    """Pressure (hPa) at z_km by the halving rule."""
    return p0_hpa * fraction_above(z_km)
```

`100 * fraction_above(11.2)` returns 25.0 and `pressure_halving(5.6)` returns 500.0.

Chapter 4

Reading the graph, and oxygen at altitude

This chapter answers Questions 3 and 4.

4.1 Reading values from a graph, and checking them

A graph turns a rule into a tool: once the curve of Figure 3.2 is drawn, any height can be looked up without calculation. To read a value:

Step 1. Find the height on the vertical axis (convert metres to kilometres first: 8848 m = 8.848 km).

Step 2. Draw a horizontal line across to the curve.

Step 3. From the meeting point, draw a vertical line down to the percentage axis and up to the pressure axis, and read both.

A hand-drawn curve and a ruler give two significant figures at best. The exact value comes from Eq. (3.4): divide the height by 5.6 km to get the number of halvings x, then $f = 2^{-x}$. On a calculator, 2^{-x} is entered as `2 ^ (-x)` or computed as $e^{-0.6931\,x}$.

Worked example: the summit of Everest, 8848 m

Number of halvings: $x = 8.848/5.6 = 1.580$. Fraction above: $f = 2^{-1.580} = 0.3345$, so **33.4 %** of the atmosphere is above the summit. Pressure: $1000 \times 0.3345 =$ **334 hPa**, Eq. (3.6). Graph check: 8.8 km lies between 5.6 km (50 %) and 11.2 km (25 %), closer to 11.2, so the answer must be between 25 and 50 %, nearer 25. It is.

4.2 Question 3: five heights

The question

Using the graph you just drew, estimate the percentage of atmosphere above and the total pressure for the following heights: Portstewart strand (0 m); the top of Mount Everest (8848 m); the cruising altitude of a jumbo jet (12 km); Machu Picchu, Peru (2430 m); the world-record hot-air balloon (21 027 m).

Answer to Question 3

Portstewart strand: **100 %, 1000 hPa**. Everest: **33.4 %, 334 hPa**. Jumbo jet at 12 km: **22.6 %, 226 hPa**. Machu Picchu: **74.0 %, 740 hPa**. Record balloon: **7.4 %, 74 hPa**. (Values read from a hand-drawn graph within about ±2 percentage points are fine.)

Check against measurement. On the summit of Everest the barometric pressure has been measured: 253 mmHg, which is 337 hPa (West et al., 1983), within 1 % of the rule's 334 hPa. The standard atmosphere gives only 314 hPa there, because it describes a cooler, mid-latitude column;

Table 4.1. Question 3 by the halving rule, Eqs. (3.4) and (3.6). The last column gives the U.S. Standard Atmosphere for comparison (NOAA et al., 1976).

Place	Height (km)	Halvings x	% above	Pressure (hPa)	Std. atm. (hPa)
Portstewart strand	0	0	100	1000	1013
Machu Picchu	2.430	0.434	74.0	740	753
Everest summit	8.848	1.580	33.4	334	314
Jumbo jet	12.000	2.143	22.6	226	193
Record balloon	21.027	3.755	7.4	74	47

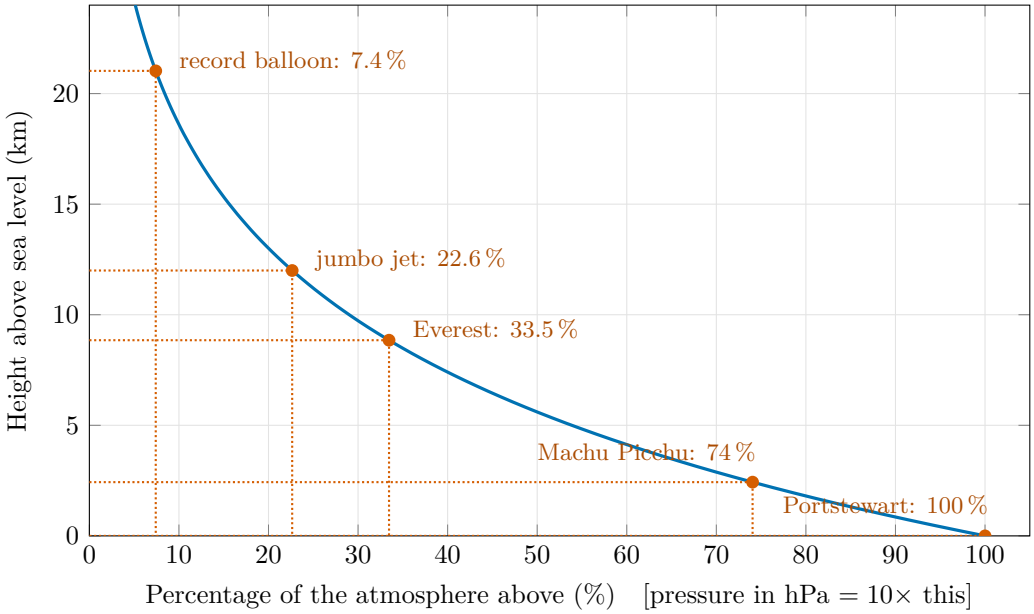


Figure 4.1. Question 3 read from the graph. For each place, go across from its height to the curve and down to the percentage axis; the pressure is ten times the percentage.

Everest stands at 28° N, where the warm air column holds the pressure up at height (West, 1996). For the jet and the balloon, high in the cold upper air, the rule overestimates the pressure, for the reason given in § 3.6.

4.3 What “21 % oxygen” means

Dry air is a mixture of gases: by volume about 78 % nitrogen, 21 % oxygen and 0.9 % argon, with carbon dioxide and other trace gases making up the rest; precise values are 78.08, 20.95 and 0.93 % (Wallace and Hobbs, 2006, ch. 1). “By volume” means by number of molecules: of every hundred molecules of dry air, about 21 are oxygen. Up to about 80 km the air is stirred so thoroughly that these proportions stay the same at every height. So the air on Everest is still 21 % oxygen. What changes is how many molecules there are in each breath.

4.4 Partial pressure

In a mixture of gases each gas contributes to the total pressure in proportion to its share of the molecules. The contribution of one gas is its **partial pressure**, and the partial pressures add up

to the total (Dalton's law). For oxygen

$$p_{O_2} = x_{O_2}\, p = 0.21\, p. \tag{4.1}$$

where

p_{O_2}	partial pressure of oxygen (hPa)	calculated
x_{O_2}	fraction of the molecules that are oxygen	0.21, rounded
p	total air pressure (hPa)	from Question 3

At sea level $p_{O_2} = 0.21 \times 1000 = 210\,\text{hPa}$ and $p_{N_2} = 0.78 \times 1000 = 780\,\text{hPa}$. The partial pressure of oxygen is what matters to the body: it pushes oxygen from the lungs into the blood. Halve the total pressure and you halve the oxygen partial pressure, even though the air is still 21 % oxygen. The same idea of a partial pressure returns for water vapour in Part III, as the vapour pressure e (§ 21.2).

4.5 Question 4: oxygen at five heights

The question

The partial pressure of each gas drops at the same rate as the total atmospheric pressure. Use your answers to Question 3 to first calculate the partial pressure of oxygen at different elevations. Then use your answer to calculate the percentage of oxygen available compared to the atmosphere at sea level.

Method.

Step 1. Partial pressure: multiply each pressure from Question 3 by 0.21, Eq. (4.1).

Step 2. Percentage: the second part of the question can be read in two ways, and the two readings give different numbers (Table 4.2).

- *(a) Oxygen as a percentage of the sea-level atmosphere:* divide the oxygen partial pressure by the sea-level *total* pressure, 1000 hPa.
- *(b) Oxygen available compared with sea level:* divide the oxygen partial pressure by the sea-level *oxygen* partial pressure, 210 hPa. This says how much oxygen each breath brings compared with a breath at the coast.

Table 4.2. Question 4. Partial pressure of oxygen, Eq. (4.1), and the two percentages.

Place	**Pressure (hPa)**	p_{O_2} **(hPa)**	**(a)** $p_{O_2}/1000$	**(b)** $p_{O_2}/210$
Portstewart strand	1000	210	21.0 %	100 %
Machu Picchu	740	155.5	15.5 %	74 %
Everest summit	334	70.2	7.0 %	33 %
Jumbo jet	226	47.6	4.8 %	23 %
Record balloon	74	15.6	1.6 %	7.4 %

Answer to Question 4

Partial pressure of oxygen: Portstewart **210 hPa**; Machu Picchu **155 hPa**; Everest **70 hPa**; jumbo jet **48 hPa**; record balloon **16 hPa**. As a percentage of the sea-level atmosphere: 21, 15.5, 7.0, 4.8 and 1.6 %. Compared with the oxygen available at sea level: 100, 74, 33, 23 and 7 %. On Everest each breath brings only a third of the oxygen it brings at sea level.

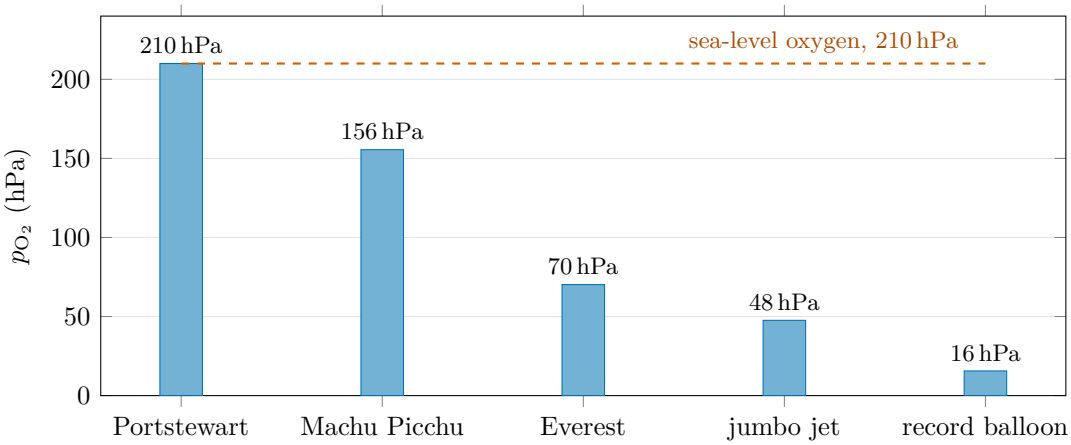


Figure 4.2. Question 4: the partial pressure of oxygen at the five places. The air is 21 % oxygen at all of them; what falls is the number of molecules in each breath, in step with the total pressure (Table 4.2).

Two ways to give the percentage

The second reading answers "how much oxygen is available compared with sea level" and is the more useful for explaining breathlessness. It is simply the percentage of the atmosphere above you from Question 3: since the oxygen fraction does not change with height, the available oxygen falls in exactly the same proportion as the total pressure.

4.6 Why it matters

- **Machu Picchu** (about three-quarters of sea-level oxygen): most visitors notice breathlessness on climbing stairs; the body adjusts within days by breathing faster and making more red blood cells.
- **Everest** (a third): close to the limit of human survival without bottled oxygen. Climbers acclimatise for weeks, and most use supplementary oxygen near the summit (West et al., 1983).
- **A jet at 12 km** (under a quarter): a person would lose consciousness within minutes. Airliner cabins are therefore pressurised, to roughly the pressure found at about 2 km, and oxygen masks drop if the pressure is lost.
- **The record balloon at 21 km** (7 %): no human could breathe this air; balloonists wear pressure suits or ride in sealed gondolas.

A question that comes up

If the air is still 21 % oxygen at the top of Everest, why is it harder to breathe? Because "21 %" is a proportion, not an amount. At the summit each lungful of air contains only a third as many molecules of every gas as at sea level, so it delivers a third as much oxygen. The lungs move oxygen into the blood in proportion to its partial pressure, 70 hPa instead of 210.

4.7 Try it yourself: Practice Set 4

Practice Set 4: reading heights and partial pressures

Use the halving rule and an oxygen fraction of 0.21.

4.1 Ben Nevis is 1345 m high. What are the pressure and the oxygen partial pressure at the summit?

4.2 Kilimanjaro is 5895 m high. Find the pressure, the oxygen partial pressure, and the oxygen available compared with sea level.

4.3 On a summit the pressure is 600 hPa. What percentage of the air molecules there are oxygen? What is the oxygen partial pressure? At what height is the summit, by the rule?

4.4 What is the partial pressure of nitrogen on the summit of Everest?

Answers to Practice Set 4

4.1 $x = 1.345/5.6 = 0.240$; $f = 2^{-0.240} = 0.847$; pressure 847 hPa; $p_{O_2} = 0.21 \times 847 = 178$ hPa.

4.2 $x = 5.895/5.6 = 1.053$; $f = 0.482$; pressure 482 hPa; $p_{O_2} = 101$ hPa; 48 % of the sea-level oxygen.

4.3 Still 21 % (the proportion does not change). $p_{O_2} = 0.21 \times 600 = 126$ hPa. Height: $2^{-z/5.6} = 0.6$, so $z = 5.6 \times \log_2(1/0.6) = 5.6 \times 0.737 = 4.1$ km.

4.4 $0.78 \times 334 = 261$ hPa.

4.8 Oxygen in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def partial_pressure(p_hpa, gas_fraction=0.21):
    """Partial pressure (hPa) of a gas that makes up gas_fraction of the molecules."""
    return gas_fraction * p_hpa
```

`partial_pressure(pressure_halving(12))` returns 47.6, the jet's oxygen partial pressure in hPa.

Chapter 5

Temperature with height: the standard atmosphere

This chapter answers Question 5 and introduces the layers of the atmosphere used in Questions 6 to 11.

5.1 Temperature and the lapse rate

Temperature measures how fast the molecules of the air are moving: the warmer the air, the faster they move. Unlike pressure, temperature does not fall steadily all the way up. It falls, then rises, then falls and rises again, and those reversals divide the atmosphere into layers (§ 5.4).

The rate at which the temperature falls with height is the **lapse rate**:

$$\Gamma = -\frac{\Delta T}{\Delta z} = \frac{T_{\text{lower}} - T_{\text{upper}}}{z_{\text{upper}} - z_{\text{lower}}}. \tag{5.1}$$

where

Γ	lapse rate ($^\circ\text{C}\,\text{km}^{-1}$, the same as $\text{K}\,\text{km}^{-1}$)	calculated
ΔT	change in temperature going up ($^\circ$C)	from a table or graph
Δz	change in height (km)	from a table or graph

The minus sign makes the lapse rate *positive* when the temperature falls with height, which is the usual case in the lowest layer. A negative lapse rate means the temperature *rises* with height: an **inversion** (§ 6.3). A lapse rate of zero means the temperature does not change: an **isothermal** layer.

5.2 The standard atmosphere

Real temperature profiles change from day to day and from place to place. For aircraft instruments, engineering and comparison, meteorologists use an agreed average: the **standard atmosphere**. Near the ground it is defined by (NOAA et al., 1976):

$$T(z) = \begin{cases} 15 - 6.5\,z & 0 \le z \le 11 \text{ km}, \\ -56.5 & 11 \le z \le 20 \text{ km}, \end{cases} \tag{5.2}$$

where

$T(z)$	temperature at height z ($^\circ$C)	calculated
z	height above sea level (km)	
15	sea-level temperature ($^\circ$C)	288.15 K
6.5	the standard ("normal") lapse rate ($^\circ\text{C}\,\text{km}^{-1}$)	agreed value

At sea level the standard pressure is 1013.25 hPa and the standard temperature 15 °C. The temperature falls by 6.5 °C for every kilometre up to 11 km, the standard **tropopause**, where it reaches −56.5 °C, and then stays constant.

5.3 Question 5: the standard atmosphere

The question

In the troposphere temperature decreases by 6.5 °C per km in the lowest 11 km; this is called the average or normal lapse rate. Based on this average lapse rate, work out the temperature of the "standard atmosphere" at 2000, 4000, 6000, 8000 and 10 000 m. You are given 15 °C at sea level, 8.5 °C at 1000 m, and −56.5 °C at 11 000, 14 000 and 16 000 m.

Method. Use Eq. (5.2). For each height up to 11 km, convert to kilometres and subtract $6.5 \times z$ from 15 °C. Above 11 km the temperature stays at −56.5 °C.

$$T(2\text{ km}) = 15 - 6.5 \times 2 = 15 - 13 = 2\ ^\circ\text{C}, \qquad T(8\text{ km}) = 15 - 6.5 \times 8 = 15 - 52 = -37\ ^\circ\text{C}.$$

Table 5.1. The standard atmosphere. Values in bold are the answers to Question 5; the others are given in the question.

Height (m)	0	1000	2000	4000	6000	8000	10 000	11 000	14 000	16 000
T (°C)	15	8.5	**2**	**−11**	**−24**	**−37**	**−50**	−56.5	−56.5	−56.5

Answer to Question 5

At 2000, 4000, 6000, 8000 and 10 000 m the standard temperatures are **2, −11, −24, −37 and −50** °C. At 11 000 m the temperature reaches −56.5 °C and stays there to 16 000 m (Table 5.1).

Check. The given values agree with the rule: $15 - 6.5 \times 1 = 8.5$ °C at 1000 m and $15 - 6.5 \times 11 = -56.5$ °C at 11 000 m. Each 2 km step lowers the temperature by 13 °C.

5.4 The layers of the atmosphere

Figure 5.1 extends the standard atmosphere to 100 km. The turning points of the temperature (the *pauses*) divide it into four layers.

Troposphere (surface to about 11 km). Temperature falls with height, on average by 6.5 °C per km. The air is heated mainly from below: sunlight passes largely through it and warms the ground, which then warms the air (Part II, § 17.7). Rising air also cools as it expands (Chapter 8). About three-quarters of the air and nearly all the water vapour are here, and almost all weather happens here.

Stratosphere (about 11 to 50 km). Temperature is constant at first, then rises with height. The warming comes from ozone, which absorbs the Sun's ultraviolet radiation (Part II, § 12.4). Warm air sitting on cooler air is very stable, so the stratosphere is calm and layered.

Mesosphere (about 50 to 85 km). Temperature falls again, to the coldest values in the atmosphere, below −85 °C at the mesopause.

Thermosphere (above about 90 km). Temperature rises steeply, as the few molecules there absorb the most energetic solar radiation.

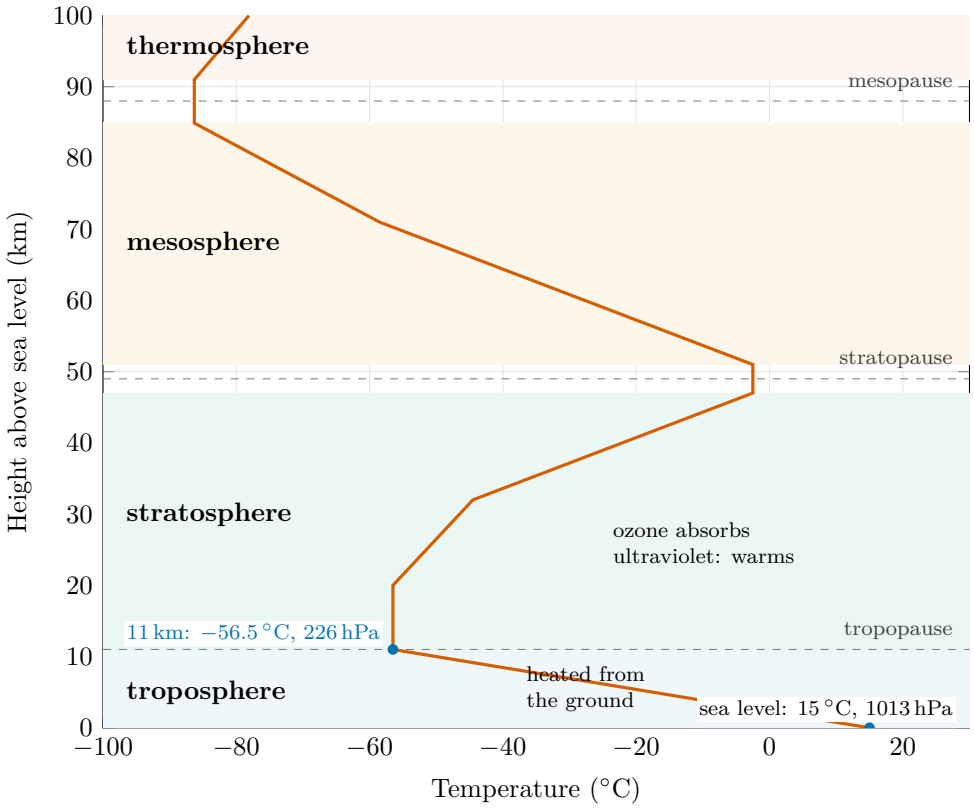


Figure 5.1. The U.S. Standard Atmosphere from the ground to 100 km (NOAA et al., 1976). Temperature falls through the troposphere, is constant and then rises through the stratosphere, falls again through the mesosphere, and rises in the thermosphere. The shaded bands are the layers; the dashed lines are the pauses between them.

A question that comes up

Is the thermosphere really hot?
Its molecules move very fast, which is what a high temperature means, and in sunlight the upper thermosphere reaches several hundred degrees. But they are so few that they would hardly heat anything: an unprotected thermometer there would lose more by radiating than it gained from the rare collisions, and would read very cold. Temperature and heat are not the same thing.

A question that comes up

Why does the troposphere get colder going up, when it is closer to the Sun?
The extra closeness is negligible (a few kilometres in 150 million). What matters is where the heating happens. Sunlight passes mostly through the air and is absorbed by the ground, so the troposphere is heated from below, like a pan of water on a hob, and is warmest at the bottom.

5.5 Try it yourself: Practice Set 5

Practice Set 5: lapse rates and the standard atmosphere

5.1 The temperature falls from 20 °C at the ground to −10 °C at 5 km. What is the lapse rate?

5.2 What is the standard-atmosphere temperature at 3 km? At the summit of Everest (8848 m)?

5.3 At what height does the standard atmosphere reach 0 °C?

5.4 Convert −56.5 °C to kelvin.

5.5 Over a layer from 1.2 to 1.6 km the temperature rises from −2 to 0 °C. What is its lapse rate, and what is such a layer called?

Answers to Practice Set 5

5.1 $\Gamma = (20 - (-10))/5 = 6\,^{\circ}\mathrm{C\,km^{-1}}$, Eq. (5.1).

5.2 $15 - 6.5 \times 3 = -4.5\,^{\circ}\mathrm{C}$; $15 - 6.5 \times 8.848 = -42.5\,^{\circ}\mathrm{C}$, Eq. (5.2).

5.3 $15 - 6.5\,z = 0$ gives $z = 15/6.5 = 2.31$ km.

5.4 $-56.5 + 273.15 = 216.65$ K.

5.5 $\Gamma = (-2 - 0)/(1.6 - 1.2) = -5\,^{\circ}\mathrm{C\,km^{-1}}$: negative, so the temperature rises with height. It is an inversion.

5.6 The standard atmosphere in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def standard_temperature(z_km):
    """Standard-atmosphere temperature (deg C): 15 - 6.5 z up to 11 km, then constant."""
    return 15.0 - 6.5 * min(z_km, 11.0)
```

`standard_temperature(8)` returns −37.0 and `standard_temperature(14)` returns −56.5.

Chapter 6

Two soundings: inversions and the tropopause

This chapter answers Questions 6, 7 and 8.

6.1 Measuring the atmosphere in the vertical

A **radiosonde** is a small instrument package carried up by a helium or hydrogen balloon. As it rises at about $5\,\mathrm{m\,s^{-1}}$ it measures pressure, temperature and humidity and sends them to the ground by radio; tracking its drift gives the wind. The balloon expands as the pressure falls and bursts at around 30 km, which is why the soundings in this chapter stop near that height. Radiosondes are launched at hundreds of stations worldwide, at the same times each day, so that the measurements can be compared and fed into weather forecasts (World Meteorological Organization, 2023). The record of one ascent is called a **sounding**.

6.2 Question 6: plotting the soundings

Table 6.1. Radiosonde soundings of 21 March 1990. Heights in metres, temperatures in °C.

Hillsborough, Northern Ireland			**St John's, Newfoundland**		
p (hPa)	z (m)	T (°C)	p (hPa)	z (m)	T (°C)
1001	38	11.05	1005	140	−1.65
853	1330	−1.25	938	687	−2.05
850	1358	−1.25	700	3006	−2.85
835	1503	−1.05	612	4062	−6.65
735	2504	−7.45	500	5604	−15.85
573	4405	−15.25	458	6263	−19.85
549	4725	−18.05	400	7248	−27.05
500	5414	−22.45	307	9103	−41.05
460	6025	−25.45	250	10 446	−53.45
400	7018	−32.85	212	11 482	−62.05
282	9383	−51.05	200	11 843	−58.05
250	10 164	−49.05	191	12 141	−54.65
216	11 116	−44.65	100	16 254	−56.25
150	13 524	−49.65	88.9	17 004	−55.85
120	14 969	−53.85	73.6	18 190	−56.65
100	16 126	−56.45	50.5	20 566	−54.85
70	18 379	−58.65	30	23 834	−56.25
50	20 482	−58.45	20	26 404	−54.05
30	23 691	−60.05	18.1	27 047	−53.65
20	26 218	−56.25	10	30 828	−54.05

The question

Table 6.1 gives the pressure, height and temperature from a radiosonde measurement taken on 21 March 1990 for Hillsborough (Northern Ireland) and St John's, Newfoundland (Canada). Plot the temperature values against height for these stations and for the standard atmosphere (Table 5.1) on one graph, using different colours for the different plots.

Method.

Step 1. Axes. Height (km) up the side, from 0 to 31 km; temperature (°C) along the bottom, from −80 to +30 °C. Height goes on the vertical axis even though it is the "input": a profile drawn this way looks like the atmosphere it describes, with the ground at the bottom.

Step 2. Convert heights to kilometres (divide the metres by 1000): 1330 m is 1.33 km.

Step 3. Plot each sounding with its own colour and symbol (Table 6.1), and the standard atmosphere from Table 5.1.

Step 4. Join the points of each sounding with straight lines. A sounding is a set of measurements at separate levels; straight lines are the honest way to connect them.

Step 5. Label each line and give the graph a title and a key.

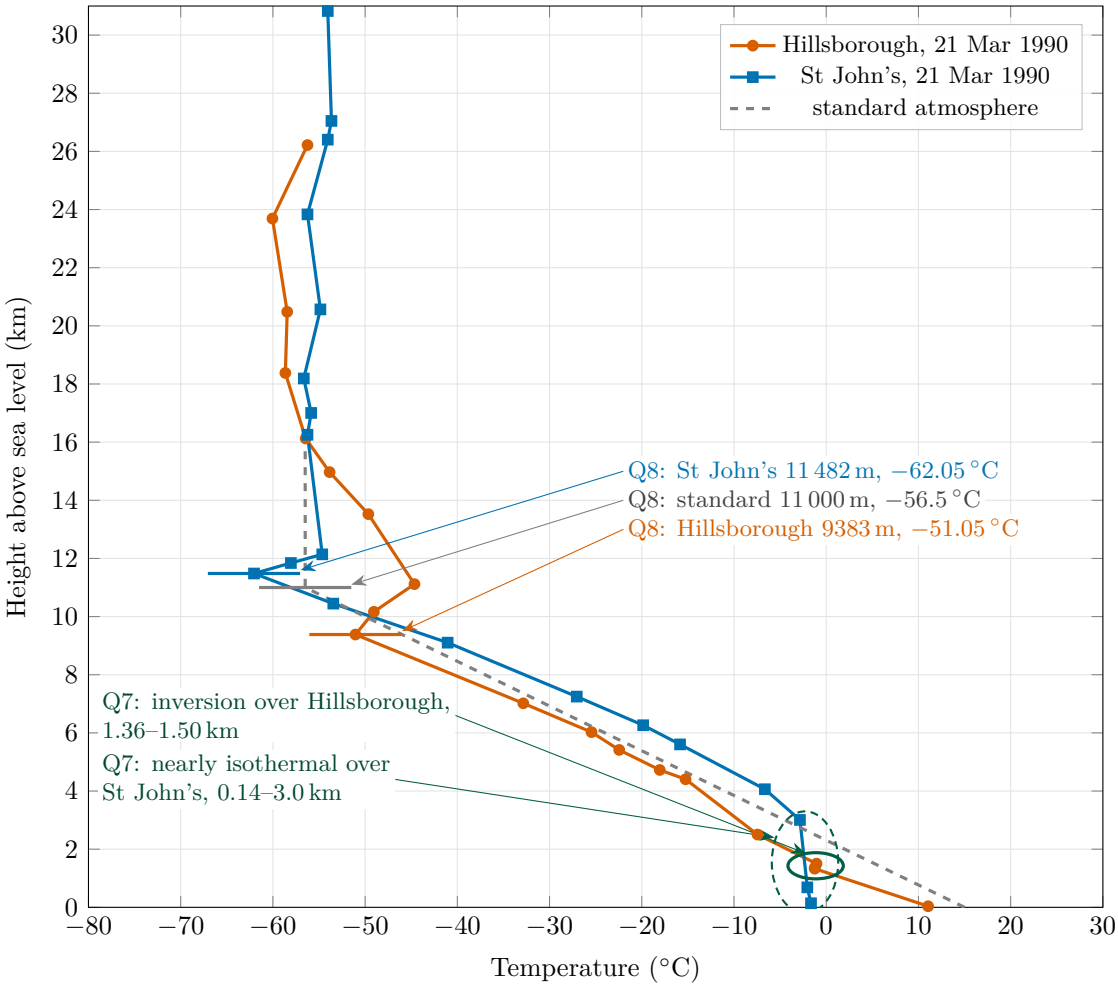


Figure 6.1. Questions 6 to 8: the two soundings of Table 6.1 and the standard atmosphere (Table 5.1), with the inversions in the troposphere circled (Question 7) and the tropopause of each profile marked (Question 8).

Answer to Question 6

The completed graph is Figure 6.1. Both soundings cool with height up to a turning point near 9 to 11 km, above which the temperature stops falling and stays between about −50 and −62 °C to 30 km. Near the ground Hillsborough is 12.7 °C warmer than St John's, but the two profiles cross at about 2 km, and between 3 and 7 km St John's is the warmer, by about 7 °C: the contrast at the surface is confined to the lowest layer (Question 14, Chapter 9).

6.3 Inversions

An **inversion** is a layer in which the temperature *rises* with height: the lapse rate, Eq. (5.1), is negative. It is called an inversion because it inverts the normal pattern of the troposphere. Inversions form in three common ways (Ahrens and Henson, 2022, ch. 3):

- **Radiation inversions:** on clear, calm nights the ground cools by radiating to the sky (Part II, § 17.8), and the air next to it cools most.
- **Subsidence inversions:** in an anticyclone, air sinking from above is compressed and warms, forming a warm layer above cooler air near the ground.
- **Frontal inversions:** warm air rides up over a wedge of colder air at a front (Part III, Chapter 29).

Inversions matter because they are very stable: they suppress rising air, trap pollution and fog beneath them, and cap the growth of clouds (Chapter 8).

6.4 Question 7: finding the inversions

The question

You will see that in some situations the temperature increases with height in the troposphere. Such layers are called inversion layers. Circle these on the graph.

Method. Go up each sounding below its tropopause and look for any step in which the temperature at the higher level is warmer than at the level below. Table 6.2 lists the lapse rate of every layer of both soundings, Eq. (5.1).

Table 6.2. The lapse rate, Eq. (5.1), of every layer of the two soundings up to 16 km. Negative lapse rates are inversions.

Layer (m)	T bottom (°C)	T top (°C)	Γ (°C km^{-1})	
Hillsborough (tropopause 9383 m)				
38–1330	11.05	−1.25	9.52	
1330–1358	−1.25	−1.25	0.00	isothermal
1358–1503	−1.25	−1.05	−1.38	inversion
1503–2504	−1.05	−7.45	6.39	
2504–4405	−7.45	−15.25	4.10	
4405–4725	−15.25	−18.05	8.75	
4725–5414	−18.05	−22.45	6.39	
5414–6025	−22.45	−25.45	4.91	

Layer (m)	T bottom (°C)	T top (°C)	Γ (°C km^{-1})	
6025–7018	−25.45	−32.85	7.45	
7018–9383	−32.85	−51.05	7.70	
9383–10 164	−51.05	−49.05	−2.56	inversion
10 164–11 116	−49.05	−44.65	−4.62	inversion
11 116–13 524	−44.65	−49.65	2.08	
13 524–14 969	−49.65	−53.85	2.91	
14 969–16 126	−53.85	−56.45	2.25	
St John's (tropopause 11 482 m)				
140–687	−1.65	−2.05	0.73	
687–3006	−2.05	−2.85	0.34	
3006–4062	−2.85	−6.65	3.60	
4062–5604	−6.65	−15.85	5.97	
5604–6263	−15.85	−19.85	6.07	
6263–7248	−19.85	−27.05	7.31	
7248–9103	−27.05	−41.05	7.55	
9103–10 446	−41.05	−53.45	9.23	
10 446–11 482	−53.45	−62.05	8.30	
11 482–11 843	−62.05	−58.05	−11.08	inversion
11 843–12 141	−58.05	−54.65	−11.41	inversion
12 141–16 254	−54.65	−56.25	0.39	

- **Hillsborough:** from 1358 m (−1.25 °C) to 1503 m (−1.05 °C) the temperature rises by 0.2 °C: a weak inversion, sitting on top of an isothermal step from 1330 to 1358 m. Below it the air cools fast, by 9.5 °C per km.
- **St John's:** no level in the troposphere is warmer than the one below, but from 140 m to 3006 m the temperature falls by only 1.2 °C in almost 3 km (0.3 to 0.7 °C per km). This deep, nearly isothermal layer is very stable and behaves like an inversion.

Answer to Question 7

Hillsborough: an inversion between about **1360 and 1500 m** (−1.25 to −1.05 °C), just above the steep cooling of the lowest 1.3 km. **St John's:** no true inversion in the troposphere, but a deep, almost **isothermal layer from the ground to about 3 km**. Both are circled in Figure 6.1.

A question that comes up

Above the tropopause the temperature rises with height too. Are those inversions? Strictly, yes: Hillsborough warms from −51 to −45 °C between 9.4 and 11.1 km, and St John's from −62 to −55 °C between 11.5 and 12.1 km. But they are in the stratosphere, where warming with height is the normal state (Figure 5.1). The question asks only about the troposphere.

6.5 The tropopause

Question 8 defines the **tropopause** as the top of the troposphere, where the temperature ceases to decrease with height. The World Meteorological Organization makes this precise: it is the lowest level at which the lapse rate falls to $2\,^{\circ}\mathrm{C\,km^{-1}}$ or less, provided the average lapse rate over the 2 km above does not exceed $2\,^{\circ}\mathrm{C\,km^{-1}}$ (Reichler et al., 2003). The second condition stops a thin inversion in the troposphere, like the one at 1.4 km over Hillsborough, from being mistaken for the tropopause. Above the tropopause the stable stratosphere begins; weather, and the tops of most clouds, stop below it.

6.6 Question 8: marking the tropopause

The question

The tropopause is defined as the top boundary of the troposphere where the temperature ceases to decrease with height. Mark this boundary on the graph and note the temperature and the height for the standard atmosphere, Hillsborough and St John's.

Method. For each profile, follow the temperature up through the troposphere and find the level above which it no longer falls. Check the WMO condition: above that level the temperature must not keep falling by more than 2 °C per km over the next 2 km.

- **Standard atmosphere:** by definition, 11 000 m and −56.5 °C (Table 5.1).
- **Hillsborough:** the temperature falls to −51.05 °C at 9383 m and then *rises* to −49.05 °C at 10 164 m and −44.65 °C at 11 116 m.
- **St John's:** the temperature falls to −62.05 °C at 11 482 m and then rises to −58.05 °C at 11 843 m.

Answer to Question 8

Standard atmosphere: **11 000 m, −56.5 °C.** Hillsborough: **9383 m, −51.05 °C.** St John's: **11 482 m, −62.05 °C.** Marked in Figure 6.1.

Applied by computer, the definition has to start its search above the lowest few kilometres. St John's has a layer from 140 m to 3 km whose temperature barely changes, and taken from the ground it would meet both conditions. The function at the end of this chapter starts at 5 km and finds the three heights above.

Check. All three lie between 9 and 12 km, the usual range for the middle latitudes (Hoinka, 1998). The higher tropopause (St John's) is also the colder one: a tropopause that sits higher has had more kilometres of cooling below it. Chapter 7 uses this to explain why the tropical tropopause is both the highest and the coldest of all.

6.7 Try it yourself: Practice Set 6

Practice Set 6: reading soundings

Use the soundings of Table 6.1.

6.1 What is the lapse rate over Hillsborough between 38 m and 1330 m? How does it compare with the standard $6.5\,^{\circ}\mathrm{C\,km^{-1}}$?

6.2 Is the layer over St John's from 9103 to 10 446 m an inversion? What is its lapse rate?

6.3 At what pressure is the tropopause over Hillsborough? Over St John's?

6.4 A sounding shows −40 °C at 9 km, −55 °C at 11 km, −54 °C at 12 km and −53 °C at 13 km. Where is the tropopause?

Answers to Practice Set 6

6.1 $(11.05 - (-1.25))/(1.330 - 0.038) = 12.30/1.292 = 9.5\,^{\circ}\mathrm{C\,km^{-1}}$: much steeper than the standard rate, and close to the rate at which rising dry air cools (§ 8.3).

6.2 No: the temperature falls from −41.05 to −53.45 °C, a lapse rate of $12.4/1.343 = 9.2\,^{\circ}\mathrm{C\,km^{-1}}$.

6.3 Hillsborough 282 hPa; St John's 212 hPa (Table 6.1).

6.4 At 11 km (−55 °C): above it the temperature rises, so the lapse rate is negative, below $2\,^{\circ}\mathrm{C\,km^{-1}}$, over the 2 km above.

6.8 Lapse rates and the tropopause in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def lapse_rate(t_lower, t_upper, z_lower_m, z_upper_m):
    """Lapse rate (deg C per km). Positive: cooler with height. Negative: an inversion."""
    return (t_lower - t_upper) / ((z_upper_m - z_lower_m) / 1000.0)
```

```
def tropopause(z_m, t_c, limit=2.0, depth_m=2000.0, start_m=5000.0):
    """WMO tropopause: the lowest level above start_m where the lapse rate falls to the limit
    or less and the mean lapse rate to every level within depth_m above stays at or below it."""
    for i in range(len(z_m) - 1):
        if z_m[i] < start_m:
            continue
        if lapse_rate(t_c[i], t_c[i + 1], z_m[i], z_m[i + 1]) > limit:
            continue
        above = [j for j in range(i + 1, len(z_m)) if z_m[j] - z_m[i] <= depth_m]
        if all(lapse_rate(t_c[i], t_c[j], z_m[i], z_m[j]) <= limit for j in above):
            return z_m[i], t_c[i]
    return None
```

`tropopause(z, t)` returns (9383, −51.05) for Hillsborough and (11482, −62.05) for St John's.

Chapter 7

Lapse rates and the height of the troposphere

This chapter answers Questions 9, 10 and 11.

7.1 Average and local lapse rates

The lapse rate of Eq. (5.1) can be taken over any layer. Taken between the ground and the tropopause it gives the **average lapse rate** of the troposphere, one number for the whole layer. Taken between neighbouring levels of a sounding it gives the **local lapse rate**, which varies a great deal: in the Hillsborough sounding it ranges from $-1.4\,^{\circ}\mathrm{C\,km^{-1}}$ (the inversion at 1.4 km) to $+9.5\,^{\circ}\mathrm{C\,km^{-1}}$ (the lowest 1.3 km), as Table 6.2 shows. The standard $6.5\,^{\circ}\mathrm{C\,km^{-1}}$ is an average of averages: over many places and many days.

7.2 Question 9: the lapse rate over Hillsborough

The question

Based on your graph and the Hillsborough sounding in Table 6.1, calculate the lapse rate for the part of the atmosphere below the tropopause for Hillsborough (have another look at Question 5 to work out how to calculate the lapse rate).

Method. Take the lowest level of the sounding and the tropopause found in Question 8 (§ 6.6), and apply Eq. (5.1).

Step 1. Lowest level: 38 m, 11.05 °C. Tropopause: 9383 m, −51.05 °C.

Step 2. Fall in temperature: $11.05 - (-51.05) = 62.10\,^{\circ}\mathrm{C}$.

Step 3. Rise in height: $9383 - 38 = 9345\,\mathrm{m} = 9.345\,\mathrm{km}$.

Step 4. Lapse rate: $\Gamma = 62.10/9.345 = 6.65\,^{\circ}\mathrm{C\,km^{-1}}$.

Answer to Question 9

$\Gamma = \dfrac{11.05 - (-51.05)}{9.383 - 0.038} = \dfrac{62.10\ ^{\circ}\mathrm{C}}{9.345\ \mathrm{km}} = \mathbf{6.65\,^{\circ}C\ per\ km}$, very close to the standard lapse rate of 6.5 °C per km.

Check. The value lies between the extremes of the local lapse rates in Table 6.2, and the straight line from the surface point to the tropopause point in Figure 6.1 runs close to the dashed standard-atmosphere line. For comparison, the same calculation for St John's gives $(-1.65 - (-62.05))/(11.482 - 0.140) = 5.33\,^{\circ}\mathrm{C\,km^{-1}}$: smaller, because its surface layer is so cold.

7.3 Question 10: a tropical profile

Table 7.1. Temperature profile for Florida.

Height (m)	0	2000	4000	6000	8000	10 000	12 000	14 000	16 000	18 000
T (°C)	27.2	15.2	6.1	−6.7	−17.9	−32.6	−50.5	−64.5	−75.6	−73.3

The question

Add the temperature curve for Florida (Table 7.1) to your graph. Where does the tropopause occur?

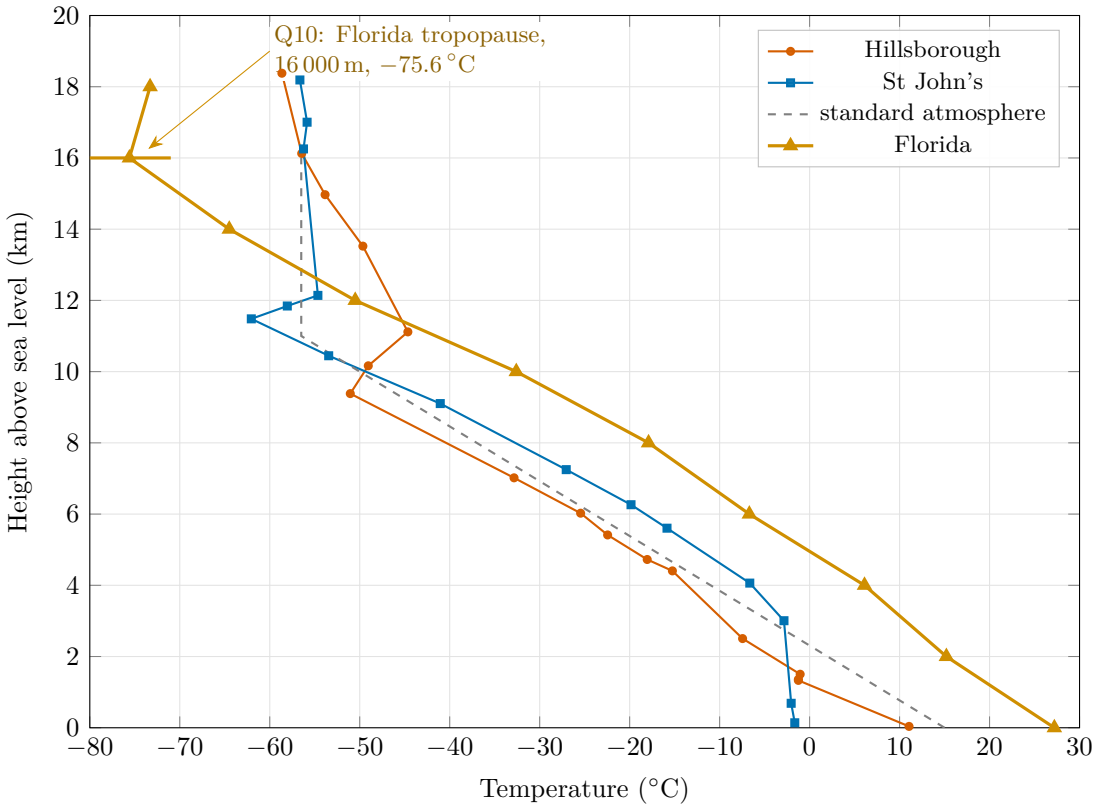


Figure 7.1. Question 10: the Florida profile (Table 7.1) added to the graph, shown here to 20 km. Florida is warmest at the ground and coldest at its tropopause, which is 5 to 7 km higher than the others.

Method. Plot Table 7.1 on the same axes (Figure 7.1) and look for the level above which the temperature stops falling: it falls all the way to −75.6 °C at 16 000 m and then rises to −73.3 °C at 18 000 m.

Answer to Question 10

Florida: the tropopause is at about **16 000 m**, at **−75.6** °C. Its average lapse rate below the tropopause is $(27.2 - (-75.6))/16 = 6.4\,^{\circ}\mathrm{C\,km^{-1}}$.

Check. The data are only every 2 km, so the true tropopause could be anywhere between about 15 and 17 km. That range is typical of the tropics and subtropics (Hoinka, 1998).

7.4 Question 11: is the troposphere the same thickness everywhere?

The question

Examining all your graphs, do you think the troposphere is the same thickness across the globe? Where do you think it is thickest and why is that?

The evidence. The four tropopauses are at 9.4 km (Hillsborough, 54° N), 11.0 km (the standard atmosphere, which represents the middle latitudes), 11.5 km (St John's, 48° N) and 16 km (Florida, about 25 to 31° N). The further from the pole, the higher the tropopause (Figure 7.2).

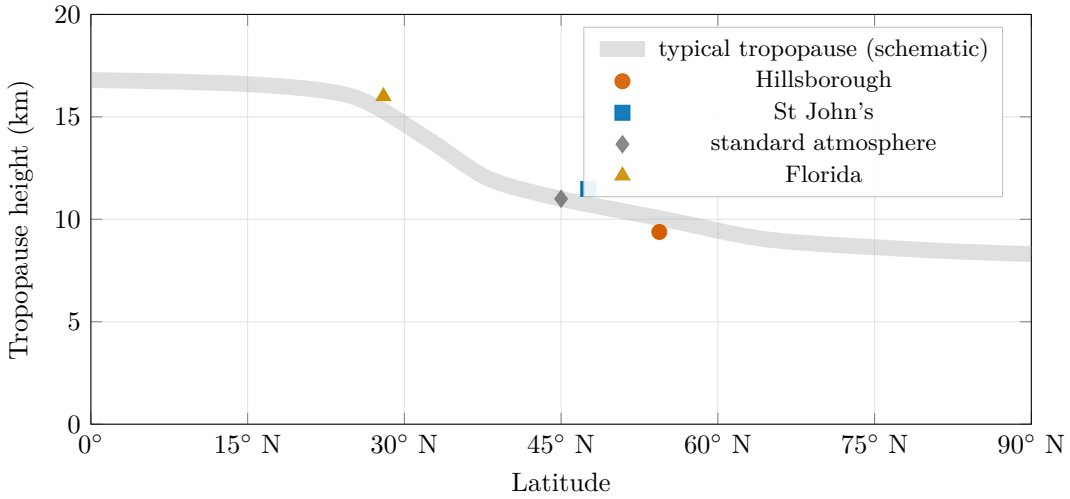


Figure 7.2. Question 11: the tropopause heights of the four profiles against latitude. The grey band shows the typical pattern (Hoinka, 1998; Wallace and Hobbs, 2006): about 16 to 17 km over the tropics, falling steeply across the subtropics and middle latitudes, to about 8 to 9 km near the poles. Florida is plotted at its middle latitude, 28° N; the standard atmosphere at 45° N.

The reasons. Three effects work together.

1. **More heating at the surface.** The tropics receive the most solar energy per square metre, because the Sun is high in the sky all year (Part II, Chapter 18). The ground and the air above it are warm.
2. **Warm air columns are taller.** Warm air is less dense than cold air at the same pressure, Eq. (3.1), so the same weight of air (the same drop in pressure) occupies a greater depth: the scale height $H = R_d T/g$ is larger, Eq. (3.3). Every pressure level, and with them the whole troposphere, sits higher above a warm surface. The same effect makes pressure surfaces slope from the warm tropics down to the cold poles, which drives the winds of Part III (Chapter 24).
3. **Deep convection.** Strong heating makes air rise in towering thunderstorm clouds that carry heat and mix the troposphere up to 15 to 18 km. The tropopause sits where this stirring from below gives way to the stable, radiatively heated stratosphere, so where convection reaches higher, the tropopause is higher.

A taller troposphere also has a *colder* top: with a similar lapse rate, the temperature keeps falling over more kilometres. That is why Florida, the warmest at the ground (27 °C), has the coldest tropopause (−75.6 °C).

Answer to Question 11

No. The troposphere is **thickest over the tropics** (about 16 to 17 km; 16 km in the Florida profile) and **thinnest near the poles** (about 8 to 9 km), with middle latitudes in between (9.4 to 11.5 km in the soundings). The tropics receive the most solar heating; warm air is less dense, so a warm air column is taller; and strong convection mixes warm air to great heights before the stable stratosphere begins. The tropopause is also higher in summer than in winter, for the same reasons.

A question that comes up

The coldest air in the troposphere is over the equator? That sounds backwards. At the ground the tropics are the warmest places on Earth. But high up, at 16 to 17 km, the tropical tropopause is colder than the tropopause anywhere else, often below $-75\,^\circ\mathrm{C}$, because it is so far above the surface. Near the poles the tropopause is low and so less cold: around -50 to $-60\,^\circ\mathrm{C}$.

7.5 Try it yourself: Practice Set 7

Practice Set 7: lapse rates

7.1 What is the lapse rate in the lowest 2 km of the Florida profile? Between 10 and 12 km?

7.2 A tropical sounding has $28\,^\circ\mathrm{C}$ at the ground and an average lapse rate of $6.5\,^\circ\mathrm{C\,km^{-1}}$ up to its tropopause at 16 km. What is the tropopause temperature?

7.3 A polar sounding has $-20\,^\circ\mathrm{C}$ at the ground and the tropopause at 8.5 km with $-55\,^\circ\mathrm{C}$. What is its average lapse rate?

7.4 By how much is the scale height over Florida (mean tropospheric temperature about $-24\,^\circ\mathrm{C}$) larger than over a polar station (about $-40\,^\circ\mathrm{C}$)?

Answers to Practice Set 7

7.1 $(27.2 - 15.2)/2 = 6.0\,^\circ\mathrm{C\,km^{-1}}$; $(-32.6 - (-50.5))/2 = 8.95\,^\circ\mathrm{C\,km^{-1}}$.

7.2 $28 - 6.5 \times 16 = 28 - 104 = -76\,^\circ\mathrm{C}$.

7.3 $(-20 - (-55))/8.5 = 35/8.5 = 4.1\,^\circ\mathrm{C\,km^{-1}}$.

7.4 $H = R_d T/g$: $287 \times 249/9.81 = 7.29$ km against $287 \times 233/9.81 = 6.82$ km, about 0.47 km (7 %) larger. Every pressure level therefore sits about 7 % higher above the ground in the warm column: 500 hPa, for example, at about 5.1 km instead of 4.7 km.

Chapter 8

Pressure aloft, buoyancy and stability

This chapter answers Questions 12 and 13.

8.1 Question 12: equal steps, unequal drops

The question

In St John's, how much is the pressure drop between 140 m and 3006 m? How much is the pressure drop between 23 834 m and 26 404 m? Seeing that the difference in altitude is almost the same, why is the difference in pressure so dissimilar?

The two drops. From Table 6.1 (St John's):

$$\begin{aligned} \text{lower layer:} &\quad 1005 - 700 = 305 \text{ hPa} \quad \text{over} \quad 3006 - 140 = 2866 \text{ m}, \\ \text{upper layer:} &\quad 30 - 20 = 10 \text{ hPa} \quad \text{over} \quad 26\,404 - 23\,834 = 2570 \text{ m}. \end{aligned}$$

Look at fractions, not amounts. In the lower layer the pressure falls to $700/1005 = 0.70$ of its starting value, a fall of 30 %; in the upper layer it falls to $20/30 = 0.67$, a fall of 33 %. The *fractional* drops over these similar depths are almost the same. That is exactly what exponential decay means (§ 3.3): each kilometre removes the same *fraction* of the air above, so where there is a lot of air (near the ground) the same fraction is a large amount, and where there is little (at 24 km, only 3 % of the atmosphere is above) it is a tiny amount. Plotted on a logarithmic pressure axis, an exponential becomes a straight line (Figure 8.1).

The data even measure the scale height. Rearranging Eq. (3.3) for two levels gives the scale height of the layer between them (Appendix A.2.1):

$$H = \frac{z_2 - z_1}{\ln(p_1/p_2)}. \tag{8.1}$$

where

H	scale height of the layer (km)	calculated
$z_1,\ z_2$	heights of the bottom and top of the layer (km)	from the sounding
$p_1,\ p_2$	pressures at those heights (hPa)	from the sounding

For the lower layer $H = 2.866/\ln(1005/700) = 2.866/0.362 = 7.9$ km; for the upper layer $H = 2.570/\ln(30/20) = 2.570/0.405 = 6.3$ km. By Eq. (3.3), $H = R_d T/g$, so these correspond to temperatures of $gH/R_d = 271$ K (-2 °C) and 218 K (-55 °C): exactly the temperatures measured in those layers. The upper layer is colder, so its air is denser for its pressure and its pressure falls off over a shorter distance.

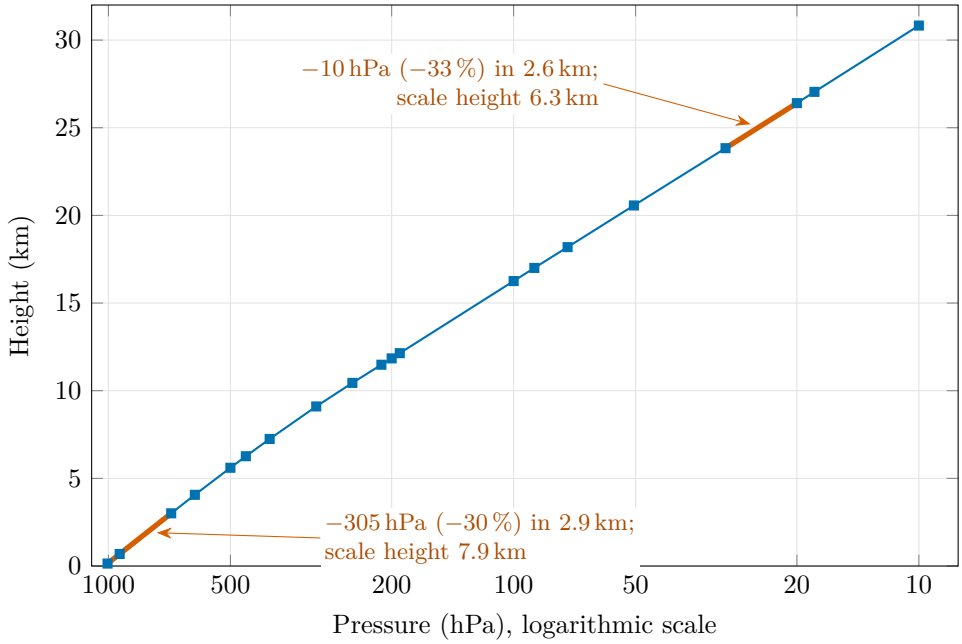


Figure 8.1. Question 12: the St John's sounding with pressure on a logarithmic axis (decreasing to the right, so that height and falling pressure go the same way). An exponential decay is a straight line here; the slope is steeper (shorter scale height) where the air is colder.

Answer to Question 12

Between 140 and 3006 m the pressure drops by **305 hPa**; between 23 834 and 26 404 m by only **10 hPa**. Pressure decreases **exponentially** with height: each step up removes about the same *fraction* of the air above (30 % and 33 % here), not the same amount. Near the ground there is a great deal of air above, compressed by its own weight, so the same fraction is a large drop; at 24 km only about 3 % of the atmosphere remains above, so the drop is tiny.

8.2 "Cold air sinks"? Density depends on pressure too

Everyday experience says cold air sinks: cold draughts collect at the floor, frost forms in hollows. That is true when cold and warm air are side by side at the *same pressure*, because by the gas law, Eq. (3.1), $\rho = p/(R_d T)$ and lower T means higher ρ. But in the vertical the pressure changes enormously, and pressure wins. Using the Hillsborough sounding:

$$\text{surface } (1001\,\text{hPa},\ 11.05\,^\circ\text{C}): \quad \rho = \frac{100\,100}{287 \times 284.2} = 1.23\ \text{kg m}^{-3},$$

$$\text{tropopause } (282\,\text{hPa},\ -51.05\,^\circ\text{C}): \quad \rho = \frac{28\,200}{287 \times 222.1} = 0.44\ \text{kg m}^{-3}.$$

The cold air at the tropopause is only a third as dense as the warm air at the ground: it is lighter, not heavier, and it does not "fall through" the air below it.

8.3 What happens to air that moves up or down

The full answer needs one more idea. Air that moves vertically changes its pressure, and so its temperature. A parcel of air brought *down* is squeezed by the rising pressure, and the work done in compressing it warms it; a parcel that rises expands and cools. If no heat is exchanged with its

surroundings (the process is *adiabatic*), dry air warms or cools at a fixed rate, the **dry adiabatic lapse rate** (Wallace and Hobbs, 2006, ch. 3):

$$\Gamma_d = \frac{g}{c_p} = \frac{9.81}{1004}\ \mathrm{K\,m^{-1}} = 9.8\ ^\circ\mathrm{C\,km^{-1}}. \tag{8.2}$$

where

Γ_d	dry adiabatic lapse rate ($^\circ$C km^{-1})	9.8
g	acceleration due to gravity (m s^{-2})	9.81
c_p	specific heat of dry air at constant pressure (J kg^{-1} K^{-1})	1004, Table D.1

The derivation is in Appendix A.2.2. The same rate decides the height of cloud base in Part III: rising air cools at 9.8 °C km^{-1} until it reaches its dew point (Part III, § 22.8).

A convenient way to compare air at different heights is to ask what temperature each parcel would have if brought down adiabatically to the standard level of 1000 hPa. That temperature is the **potential temperature**:

$$\theta = T\left(\frac{1000\ \mathrm{hPa}}{p}\right)^{R_d/c_p} = T\left(\frac{1000}{p}\right)^{0.286}. \tag{8.3}$$

where

θ	potential temperature (K)	calculated
T	temperature of the parcel (K)	from the sounding
p	pressure of the parcel (hPa)	from the sounding
R_d/c_p	ratio of the gas constant to the specific heat	287/1004 = 0.286

Potential temperature does not change when air moves up or down adiabatically, so it labels a parcel the way a name does: two parcels with the same θ would have the same temperature if brought to the same level.

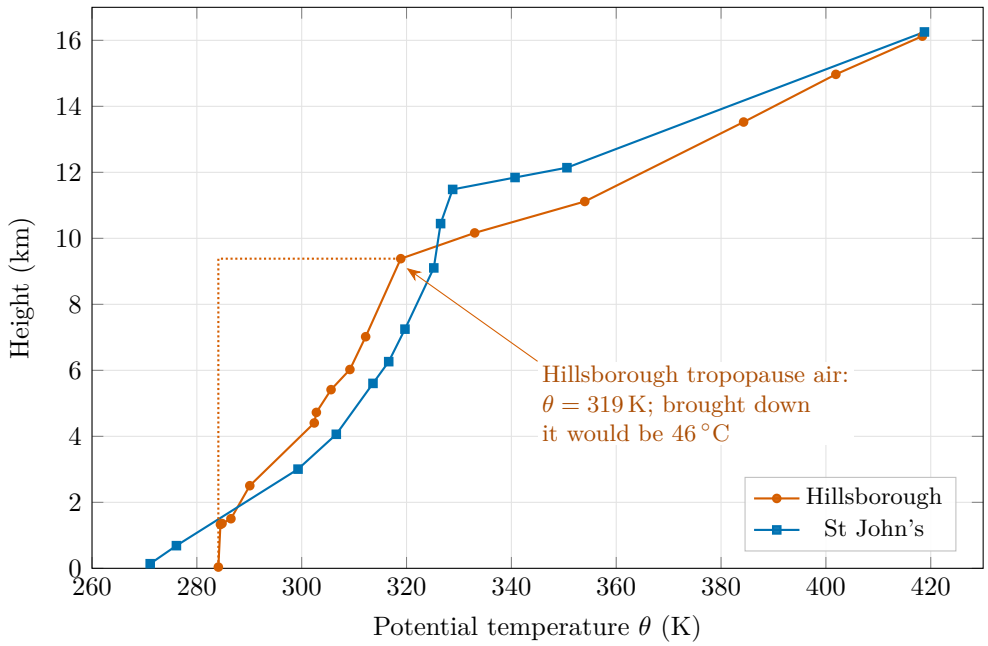


Figure 8.2. Question 13: potential temperature, Eq. (8.3), for both soundings up to 16 km. It increases with height everywhere, which means the atmosphere is stable: any parcel moved down arrives warmer, and so lighter, than the air around it. The increase is slow in the troposphere and fast in the stratosphere.

8.4 Question 13: why the cold air aloft does not sink

The question

Thinking of your previous answer, if cold air is supposed to sink, why is the colder air of the upper troposphere not flowing down to the surface?

Follow a parcel down. Take the air at the Hillsborough tropopause, $-51.05\,^\circ\mathrm{C}$ (222.1 K) at 282 hPa, and bring it down to the ground. As it sinks it is compressed and warms at $9.8\,^\circ\mathrm{C\,km^{-1}}$. Its potential temperature, Eq. (8.3), tells us where it would end up:

$$\theta = 222.1 \times \left(\frac{1000}{282}\right)^{0.286} = 222.1 \times 1.436 = 318.9\ \mathrm{K} = 45.8\ ^\circ\mathrm{C}.$$

It would arrive at the surface at about 46 °C, some 35 °C *warmer* than the 11 °C air already there. Warmer air at the same pressure is less dense, so the parcel would be buoyant and would be pushed straight back up. The cold air aloft is not "waiting to fall": it is held up because it would become warm, and light, on the way down.

This is general. Whenever the temperature in the atmosphere falls with height more slowly than $9.8\,^\circ\mathrm{C\,km^{-1}}$ (the standard $6.5\,^\circ\mathrm{C\,km^{-1}}$, or the $6.65\,^\circ\mathrm{C\,km^{-1}}$ of Question 9), the potential temperature increases with height (Figure 8.2) and the air is **stably stratified**: air displaced downwards comes back up, and air displaced upwards sinks back. The atmosphere is layered like oil on water, but the layers are sorted by potential temperature, not by temperature.

Answer to Question 13

The upper air is cold but it is at very low pressure, so it is **much less dense** than the air near the ground (0.44 against $1.23\,\mathrm{kg\,m^{-3}}$ at Hillsborough): "cold air sinks" compares air at the same pressure. If the upper air did sink, it would be **compressed and warm up** at about $9.8\,^\circ\mathrm{C}$ per km; the tropopause air over Hillsborough would reach the ground at about **46** °C, far warmer and lighter than the surface air, and would rise again. Because the temperature falls with height more slowly than air warms on descent, the atmosphere is **stably layered**: its potential temperature increases with height.

A question that comes up

Then when does air overturn?
When the temperature falls with height *faster* than $9.8\,^\circ\mathrm{C\,km^{-1}}$, a rising parcel stays warmer than its surroundings and keeps rising: the air is unstable, and convection overturns it. This happens over ground heated strongly by the Sun, where the lowest layer can exceed the dry adiabatic rate. In the Hillsborough sounding the lowest 1.3 km cools at $9.5\,^\circ\mathrm{C\,km^{-1}}$, almost exactly the dry adiabatic rate: a well-mixed layer stirred by the wind and by heating from the ground.

8.5 Try it yourself: Practice Set 8

Practice Set 8: pressure and stability

Use $R_d = 287\,\mathrm{J\,kg^{-1}\,K^{-1}}$, $c_p = 1004\,\mathrm{J\,kg^{-1}\,K^{-1}}$, $g = 9.81\,\mathrm{m\,s^{-2}}$ and kelvin $=^\circ\mathrm{C} + 273.15$.

8.1 What is the density of air at 500 hPa and $-20\,^\circ\mathrm{C}$?

8.2 What is the potential temperature of the tropopause air over St John's (212 hPa,

$-62.05\,°\mathrm{C}$)?

8.3 From the Hillsborough sounding, find the scale height of the layer from 1001 hPa (38 m) to 500 hPa (5414 m), and the temperature it corresponds to.

8.4 Air at 5414 m over Hillsborough ($-22.45\,°\mathrm{C}$) is brought down to the ground (38 m) without exchanging heat. Roughly what temperature would it have?

Answers to Practice Set 8

8.1 $\rho = 50\,000/(287 \times 253.15) = 0.688\,\mathrm{kg\,m^{-3}}$, Eq. (3.1): just over half the surface density.

8.2 $\theta = 211.1 \times (1000/212)^{0.286} = 328.8\,\mathrm{K}$ ($55.7\,°\mathrm{C}$), Eq. (8.3).

8.3 $H = 5.376/\ln(1001/500) = 5.376/0.694 = 7.74\,\mathrm{km}$, Eq. (8.1); $T = gH/R_d = 9.81 \times 7744/287 = 265\,\mathrm{K}$ ($-8\,°\mathrm{C}$), close to the average temperature of that layer.

8.4 It warms by about $9.8 \times 5.4 = 53\,°\mathrm{C}$, to about $30\,°\mathrm{C}$, Eq. (8.2): $19\,°\mathrm{C}$ warmer than the surface air, so it would rise again.

8.6 Stability in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def scale_height_from_data(p1_hpa, p2_hpa, dz_km):
    """Scale height (km) from two pressures dz_km apart."""
    return dz_km / math.log(p1_hpa / p2_hpa)
```

```
def dry_adiabatic_lapse_rate():
    """Dry adiabatic lapse rate (deg C per km): g / c_p."""
    return G / C_P * 1000.0
```

```
def potential_temperature(t_c, p_hpa):
    """Potential temperature (K) of air at t_c (deg C) and p_hpa."""
    return (t_c + 273.15) * (1000.0 / p_hpa) ** (R_D / C_P)
```

`dry_adiabatic_lapse_rate()` returns 9.77, which rounds to the $9.8\,°\mathrm{C\,km^{-1}}$ used in this chapter.

Chapter 9

Two soundings, two climates

This chapter answers Question 14.

9.1 The puzzle

On 21 March 1990 the air near the ground was at 11.05 °C over Hillsborough and −1.65 °C over St John's (Table 6.1). Hillsborough is at about 54.4° N; St John's at about 47.6° N, about 750 km closer to the equator. If latitude alone set the temperature, St John's should be the warmer.

The Sun does not explain it either. The soundings were made at the March equinox, when the noon Sun is overhead at the equator and its angle above the horizon is 90° minus the latitude (Part II, § 18.3). By Eq. (18.3), the noon solar intensity is $100\,\% \times \sin\beta$:

$$\text{Hillsborough: } 100\,\% \times \sin(35.6^\circ) = 58\,\%; \qquad \text{St John's: } 100\,\% \times \sin(42.4^\circ) = 67\,\%.$$

St John's received *more* solar energy per square metre at noon, yet was 12.7 °C colder. Something other than sunshine controls the temperature of the air near the ground.

9.2 What the soundings show

The difference is confined to the lowest layer. The two profiles cross at about 2 km, and from 3 to 7 km St John's is the warmer by about 7 °C (Figure 6.1). Over St John's the lowest 3 km is cold and almost isothermal (Question 7, § 6.4); over Hillsborough the lowest 1.3 km is warm and well mixed, cooling at nearly the dry adiabatic rate (§ 8.3). The warmth of Hillsborough, and the cold of St John's, come from the surface beneath and the air that has flowed across it.

9.3 The ocean and the winds

Two things decide the temperature of the air near the ground in winter and early spring: the temperature of the surface it lies over, and the surface it has just crossed (Lutgens et al., 2019, ch. 3).

1. **Ocean currents.** The Gulf Stream carries warm water from the Gulf of Mexico along the coast of North America and, as the North Atlantic Current, across to north-west Europe. The seas around Ireland are therefore far warmer in winter than their latitude would suggest. Newfoundland is washed instead by the Labrador Current, which brings cold water (and, in late winter and spring, sea ice and icebergs) south from the Arctic.
2. **Prevailing winds.** In the middle latitudes the winds blow mostly from the west. Air reaching Northern Ireland has crossed thousands of kilometres of relatively warm ocean and arrives mild and moist (maritime air). Air reaching St John's has usually crossed the North American continent, which in winter is snow-covered and very cold (continental polar air).
3. **Water stores heat.** Water heats and cools far more slowly than land, so the ocean keeps the air over it mild in winter and cool in summer. Downwind of an ocean, winters are mild; downwind of a continent, they are harsh.

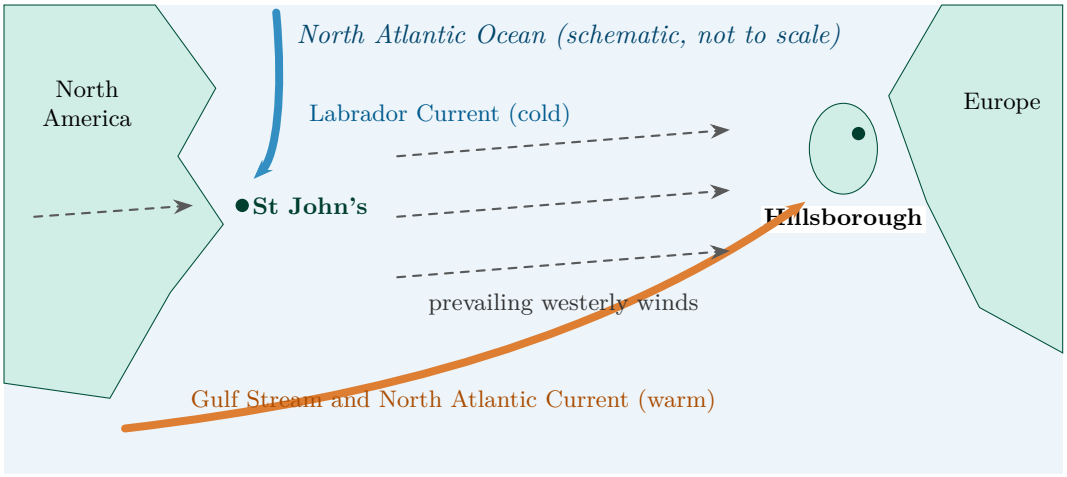


Figure 9.1. Why Hillsborough is milder than St John's in late winter (schematic). Westerly winds reach Northern Ireland after crossing ocean warmed by the Gulf Stream and North Atlantic Current; they reach Newfoundland after crossing the cold continent, and its coast is bathed by the cold Labrador Current.

9.4 Question 14: Hillsborough and St John's

The question

Compare the surface temperatures of St John's and Hillsborough. Although Northern Ireland is located at a higher latitude, why do you think the surface temperature is higher? Do you think that this difference is the same throughout the year?

Answer to Question 14

On 21 March 1990 Hillsborough was at **11.05** °C and St John's at **−1.65** °C: **12.7** °C **warmer** in Northern Ireland, although it is about 7° of latitude further north and received less sunshine at noon. Northern Ireland lies downwind of the North Atlantic, whose surface is kept warm by the **Gulf Stream and North Atlantic Current**, so the prevailing westerly winds bring it mild maritime air. St John's is bathed by the cold **Labrador Current** and its westerly winds arrive from the frozen North American continent. **The difference is not the same all year:** it is largest in winter and early spring, when the ocean is much warmer than the land and the continent is coldest, and much smaller in summer, when the continent warms up (summer days in St John's can be warmer than in Northern Ireland).

Check. The soundings agree with this explanation: the contrast is confined to the lowest kilometre or two, the layer in contact with the surface, while higher up St John's is the warmer (§ 9.2). A difference caused by latitude or by the Sun would extend through the whole troposphere.

A question that comes up

Is one day's sounding enough to say anything about climate?
No. One day shows weather; climate is the average over many years. But this day is typical of the pattern: north-west Europe has some of the mildest winters for its latitude anywhere in the world, and eastern Canada some of the coldest. The sounding illustrates a climatic contrast; the climate itself is established from long records.

9.5 Try it yourself: Practice Set 9

Practice Set 9: places and seasons

9.1 At the June solstice, what are the noon solar intensities at Hillsborough (54.4° N) and St John's (47.6° N)? Use Part II, Eqs. (18.4) and (18.3).

9.2 Bergen in Norway (60.4° N) has milder winters than many places 15° further south in eastern North America. Give two reasons.

9.3 Which of the two soundings has the more stable lowest kilometre, and why does that fit the explanation of Question 14?

Answers to Practice Set 9

9.1 Zenith angles $54.4 - 23.5 = 30.9°$ and $47.6 - 23.5 = 24.1°$; sun angles $59.1°$ and $65.9°$; intensities $\sin 59.1° = 86\,\%$ and $\sin 65.9° = 91\,\%$ per unit.

9.2 The North Atlantic Current keeps the Norwegian Sea warm and ice-free in winter, and the prevailing westerly winds bring mild maritime air from the ocean, whereas eastern North America receives air from the cold continent.

9.3 St John's: its lowest 3 km is almost isothermal (a lapse rate under $1\,^\circ\mathrm{C\,km^{-1}}$), so cold air lies trapped beneath warmer air, as over a cold surface. Hillsborough's lowest 1.3 km cools at $9.5\,^\circ\mathrm{C\,km^{-1}}$, a mixed layer over a relatively warm surface.

Chapter 10

Putting Part I together

10.1 One chain of ideas

The fourteen questions are one argument, each step resting on the one before:

1. Air has weight, and pressure is the weight of the air above. Because air is compressible, the air near the ground is squeezed by everything above it: pressure and density fall **exponentially** with height, halving roughly every 5.6 km (Questions 1–2, Chapter 3).
2. Reading that curve gives the pressure anywhere; because the proportion of oxygen is the same at all heights, its **partial pressure** falls in step, to a third on Everest (Questions 3–4, Chapter 4).
3. Temperature falls through the troposphere, on average by 6.5 °C per km, then stops falling at the **tropopause**, above which the ozone-heated stratosphere begins (Question 5, Chapter 5).
4. Real soundings follow the standard atmosphere only roughly: they contain **inversions**, and their tropopauses lie at different heights (Questions 6–8, Chapter 6).
5. The troposphere is **thickest and its top coldest over the tropics**, where heating and convection are strongest and warm air columns are tallest (Questions 9–11, Chapter 7).
6. Equal height steps remove equal *fractions* of the air, so pressure falls fast low down and slowly high up; and air that sinks warms by compression, so the cold air aloft is held up: the atmosphere is **stably layered** (Questions 12–13, Chapter 8).
7. Near the ground, the surface and the winds over it control the temperature: an ocean warmed by the Gulf Stream keeps Northern Ireland mild (Question 14, Chapter 9).

Part II explains where the heat comes from in the first place, and Part III how the differences of pressure set up here make the air move.

10.2 All the answers in one table

Table 10.1. Every answer of Part I, with the section where it is worked out.

Q	Answer	Where
1	100, 50, 25, 12.5, 6.25, 3.13, 1.56, 0.78, 0.39, 0.20 % above 0, 5.6, . . . , 50.4 km	§ 3.4
2	1000, 500, 250, 125, 62.5, 31.3, 15.6, 7.8, 3.9, 2.0 hPa; second axis = 10× the percentage	§ 3.5
3	Portstewart 100 %, 1000 hPa; Everest 33.4 %, 334 hPa; jet 22.6 %, 226 hPa; Machu Picchu 74.0 %, 740 hPa; balloon 7.4 %, 74 hPa	§ 4.2
4	p_{O_2}: 210, 70, 48, 155, 16 hPa; as % of the sea-level atmosphere 21, 7.0, 4.8, 15.5, 1.6 %; compared with sea-level oxygen 100, 33, 23, 74, 7 %	§ 4.5

Q	Answer	Where
5	2, −11, −24, −37, −50 °C at 2, 4, 6, 8, 10 km; −56.5 °C from 11 to 16 km	§ 5.3
6	graph of the two soundings and the standard atmosphere (Figure 6.1)	§ 6.2
7	Hillsborough inversion about 1360–1500 m; St John's nearly isothermal from the ground to about 3 km	§ 6.4
8	standard 11 000 m, −56.5 °C; Hillsborough 9383 m, −51.05 °C; St John's 11 482 m, −62.05 °C	§ 6.6
9	62.10/9.345 = 6.65 °C per km	§ 7.2
10	Florida tropopause about 16 000 m, −75.6 °C	§ 7.3
11	no: thickest over the tropics (16–17 km), thinnest at the poles (8–9 km); heating, warm tall columns, deep convection	§ 7.4
12	305 hPa against 10 hPa; exponential decay: equal fractions (30 and 33 %), not equal amounts	§ 8.1
13	upper air is far less dense (low pressure); sinking air warms by compression (9.8 °C per km): tropopause air would arrive at 46 °C; stable layering	§ 8.4
14	Hillsborough 12.7 °C warmer: Gulf Stream and North Atlantic Current, westerly maritime air; Labrador Current and continental air at St John's; largest contrast in winter and spring	§ 9.4

10.3 Part I on one page

The essentials

Halving rule: $f = 2^{-z/5.6}$; % above = $100f$; pressure = $1000f$ hPa — Eqs. (3.4), (3.6)
Steps of 5.6 km: 100, 50, 25, 12.5, 6.25 … % — Table 3.1
Oxygen: $p_{O_2} = 0.21\,p$; available oxygen falls in step with p — Eq. (4.1)
Lapse rate: $\Gamma = (T_{\text{lower}} - T_{\text{upper}})/(z_{\text{upper}} - z_{\text{lower}})$; negative means inversion — Eq. (5.1)
Standard atmosphere: $T = 15 - 6.5\,z$ to 11 km, then −56.5 °C — Eq. (5.2)
Tropopause: lowest level where the temperature stops falling (and stays within 2 °C km^{-1} for the 2 km above) — § 6.5
Scale height: $H = R_d T/g \approx 8$ km; from data $H = \Delta z/\ln(p_1/p_2)$ — Eqs. (3.3), (8.1)
Sinking air warms: 9.8 °C per km; $\theta = T(1000/p)^{0.286}$ — Eqs. (8.2), (8.3)

Graph habits: height always on the vertical axis; one colour and symbol per profile; straight lines between sounding points, a smooth curve for the halving rule; convert metres to kilometres before plotting.

10.4 Mistakes to avoid

- **Halving the height instead of the air.** Each 5.6 km step halves the *percentage*; the heights go up in equal steps.
- **Joining the halving-rule points with straight lines.** The rule describes a smooth exponential curve; straight segments make the readings of Question 3 wrong between the points.

- **Mixing metres and kilometres** in the lapse rate or in $z/5.6$.
- **Getting the sign of the lapse rate wrong.** Temperature falling with height is a *positive* lapse rate.
- **Taking any warm step as the tropopause.** A thin inversion low in the troposphere (Hillsborough, 1.4 km) is not the tropopause: the temperature falls again above it.
- **Saying the air aloft is "heavier" because it is colder.** At the same pressure, yes; at its own low pressure it is far lighter.

10.5 Self-test

Self-test: one line each

S1. Why does pressure fall faster near the ground than high up?

S2. Why does the air on Everest have less oxygen, though it is still 21 % oxygen?

S3. What is the difference between an inversion and the tropopause?

S4. Why is the tropical tropopause the coldest?

S5. Why does the cold air of the upper troposphere not sink?

S6. Why is Northern Ireland milder in winter than Newfoundland?

Answers to the self-test

S1. Each height step removes the same fraction of the air above, and near the ground there is far more air above (§ 8.1).

S2. Each breath holds a third as many molecules; the oxygen partial pressure is a third of that at sea level (§ 4.5).

S3. An inversion is any layer where the temperature rises with height; the tropopause is the top of the troposphere, above which the temperature stops falling for good (§ 6.5).

S4. It is the highest, so the temperature falls over the most kilometres before it stops (§ 7.4).

S5. It is far less dense than the air below, and if it sank it would warm by compression and become lighter still (§ 8.4).

S6. Westerly winds off an ocean warmed by the Gulf Stream, against a cold current and a frozen continent upwind of Newfoundland (§ 9.4).

Appendix A

Part I: formulas, derivations and symbols

A.1 Formulas

Every formula used in Part I, with the equation number where it is introduced and explained.

Quantity	Formula	Where, and what for
Ideal gas law	$p = \rho R_d T$	Eq. (3.1); density from pressure and temperature (Q13)
Hydrostatic balance	$\Delta p/\Delta z = -\rho g$	Eq. (3.2); why pressure falls with height
Exponential pressure	$p = p_0 e^{-z/H}$, $H = R_d T/g$	Eq. (3.3); scale height (Q1, Q12)
Fraction of the atmosphere above	$f = 2^{-z/5.6}$ (z in km)	Eq. (3.4); Q1, Q3
After n halvings	$f_n = 1/2^n$	Eq. (3.5); Table 3.1
Pressure from the fraction	$p = 1000\, f$ hPa	Eq. (3.6); Q2, Q3
Partial pressure of oxygen	$p_{\mathrm{O_2}} = 0.21\, p$	Eq. (4.1); Q4
Lapse rate	$\Gamma = -\Delta T/\Delta z$	Eq. (5.1); Q5, Q7–Q10
Standard atmosphere	$T = 15 - 6.5\, z$ to 11 km, then $-56.5\,^\circ$C	Eq. (5.2); Q5
Scale height from data	$H = (z_2 - z_1)/\ln(p_1/p_2)$	Eq. (8.1); Q12
Dry adiabatic lapse rate	$\Gamma_d = g/c_p = 9.8\,^\circ\mathrm{C\,km^{-1}}$	Eq. (8.2); Q13
Potential temperature	$\theta = T(1000/p)^{0.286}$	Eq. (8.3); Q13
Kelvin and Celsius	$T_\mathrm{K} = T_{^\circ\mathrm{C}} + 273.15$	§ 1.3

A.2 Derivations for the curious

A.2.1 Pressure falls exponentially with height

Start from hydrostatic balance, Eq. (3.2), written for a very thin layer, $dp/dz = -\rho g$, and replace the density using the gas law, Eq. (3.1), $\rho = p/(R_d T)$:

$$\frac{dp}{dz} = -\frac{g}{R_d T}\, p \qquad \Longrightarrow \qquad \frac{dp}{p} = -\frac{dz}{H}, \qquad H = \frac{R_d T}{g}.$$

The rate of fall of p is proportional to p itself. If T (and so H) is taken as constant, integrating from sea level ($z = 0$, $p = p_0$) to height z gives $\ln(p/p_0) = -z/H$, that is

$$p = p_0\, e^{-z/H},$$

which is Eq. (3.3). The pressure halves when $e^{-z/H} = \frac{1}{2}$, at $z_{1/2} = H \ln 2 = 0.693\, H$; the 5.6 km corresponds to $H = 8.08$ km. Writing the same law for two levels and dividing, $p_1/p_2 = e^{(z_2 - z_1)/H}$, so $\ln(p_1/p_2) = (z_2 - z_1)/H$: rearranged, this is Eq. (8.1), which gives the scale height of a layer from two lines of a sounding.

Without calculus. Divide the atmosphere into thin slabs, each Δz thick. By Eqs. (3.2) and (3.1), crossing one slab lowers the pressure by $\Delta p = \rho g\, \Delta z = p\, \Delta z/H$: the same *fraction* $\Delta z/H$ of whatever pressure there is. Removing the same fraction again and again is exactly what an exponential does.

A.2.2 The dry adiabatic lapse rate and potential temperature

The first law of thermodynamics for one kilogram of air says that heat added goes into warming the air and into the work it does as it expands: $dq = c_p\, dT - dp/\rho$. In an adiabatic process no heat is added, $dq = 0$, so

$$c_p\, dT = \frac{dp}{\rho}.$$

For air moving vertically through an atmosphere in hydrostatic balance, $dp = -\rho g\, dz$, Eq. (3.2). Substituting,

$$c_p\, dT = -g\, dz \qquad \Longrightarrow \qquad \frac{dT}{dz} = -\frac{g}{c_p} = -\frac{9.81}{1004} = -0.0098\ \mathrm{K\,m^{-1}},$$

which is Eq. (8.2): dry air cools by 9.8 °C per km as it rises and warms by the same amount as it sinks. Alternatively, dividing $c_p\, dT = dp/\rho$ by T and using $1/\rho = R_d T/p$ gives $c_p\, dT/T = R_d\, dp/p$; integrating from (T, p) to $(\theta, 1000\ \mathrm{hPa})$,

$$c_p \ln\frac{\theta}{T} = R_d \ln\frac{1000}{p} \qquad \Longrightarrow \qquad \theta = T\left(\frac{1000}{p}\right)^{R_d/c_p},$$

which is Eq. (8.3), with $R_d/c_p = 287/1004 = 0.286$ (Wallace and Hobbs, 2006, ch. 3).

A.3 Symbols

Table A.1. The symbols used in Part I. Each equation's own *where* list repeats the symbols it uses.

Symbol	Meaning	Unit	First used
c_p	specific heat of dry air at constant pressure	$\mathrm{J\,kg^{-1}\,K^{-1}}$	Eq. (8.2)
f	fraction of the atmosphere above a height	–	Eq. (3.4)
g	acceleration due to gravity	$\mathrm{m\,s^{-2}}$	Eq. (3.2)
Γ	lapse rate	$\mathrm{°C\,km^{-1}}$	Eq. (5.1)
Γ_d	dry adiabatic lapse rate	$\mathrm{°C\,km^{-1}}$	Eq. (8.2)
H	scale height	km	Eq. (3.3)
p	pressure	hPa or Pa	Eq. (3.1)

Symbol	Meaning	Unit	First used
p_0	pressure at sea level	hPa	Eq. (3.3)
p_{O_2}	partial pressure of oxygen	hPa	Eq. (4.1)
R_d	gas constant for dry air	$\mathrm{J\,kg^{-1}\,K^{-1}}$	Eq. (3.1)
ρ	density of the air	$\mathrm{kg\,m^{-3}}$	Eq. (3.1)
T	temperature	K or °C	Eq. (3.1)
θ	potential temperature	K	Eq. (8.3)
x_{O_2}	fraction of the molecules that are oxygen	–	Eq. (4.1)
z	height above sea level	m or km	Eq. (3.2)

A.4 About the figures and the data

Every figure in Part I was drawn for it in the document itself with TikZ and pgfplots, from the data and the equations in the text.

- The **halving-rule curves** (Figures 3.2 and 4.1) are Eq. (3.4); the dashed comparison curve and the layered profile of Figure 5.1 are the U.S. Standard Atmosphere (NOAA et al., 1976).
- The **soundings** (Figures 6.1, 7.1 and 8.1) plot the values of Tables 6.1 and 7.1, joined by straight lines.
- The **potential temperatures** (Figure 8.2) are computed from the soundings with Eq. (8.3), and the **oxygen partial pressures** (Figure 4.2) with Eq. (4.1).
- The **hydrostatic slab** (Figure 3.1) is a diagram of the forces in Eq. (3.2).
- The **tropopause band** of Figure 7.2 and the **ocean-current sketch** of Figure 9.1 are schematic: they show the pattern, not measurements.

The soundings are single ascents on a single day; they illustrate the structure of the atmosphere and should not be read as climatological averages (§ 9.4).

Appendix D

Constants, units and conversions

D.1 Constants

Each part works with the rounded values below. The precise values are given for reference.

Table D.1. Constants used in the book.

Quantity	Value used	Precise value	Source	Part
Acceleration due to gravity g	$9.81\,\mathrm{m\,s^{-2}}$	$9.80665\,\mathrm{m\,s^{-2}}$ (standard)	NOAA et al. (1976)	I, III
Gas constant for dry air R_d	$287\,\mathrm{J\,kg^{-1}\,K^{-1}}$	$287.05\,\mathrm{J\,kg^{-1}\,K^{-1}}$	Wallace and Hobbs (2006)	I, III
Specific heat of dry air c_p	$1004\,\mathrm{J\,kg^{-1}\,K^{-1}}$	$1004\,\mathrm{J\,kg^{-1}\,K^{-1}}$	Wallace and Hobbs (2006)	I
Standard sea-level pressure	1000 hPa (Part I)	1013.25 hPa	NOAA et al. (1976)	I
Standard sea-level temperature	15 °C	288.15 K	NOAA et al. (1976)	I
Stefan–Boltzmann constant σ	$5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$	5.670374×10^{-8}	Tiesinga et al. (2021)	II
Wien's constant C	$2898\,\mu\mathrm{m\,K}$	$2897.77\,\mu\mathrm{m\,K}$	Tiesinga et al. (2021)	II
Planck's constant h	–	$6.62607015 \times 10^{-34}\,\mathrm{J\,s}$ (exact)	Tiesinga et al. (2021)	II
Speed of light c	–	$2.99792458 \times 10^{8}\,\mathrm{m\,s^{-1}}$ (exact)	Tiesinga et al. (2021)	II
Boltzmann's constant k_B	–	$1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ (exact)	Tiesinga et al. (2021)	II
Solar constant S	$1368\,\mathrm{W\,m^{-2}}$	$1361\,\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)	II
Earth's axial tilt	23.5°	23.44°	Stull (2017)	II
Earth's rotation rate Ω	$7.292 \times 10^{-5}\,\mathrm{s^{-1}}$	$7.292 \times 10^{-5}\,\mathrm{s^{-1}}$	Holton and Hakim (2013)	III
Kelvin–Celsius offset	273.15 (273.16 in Part II)	273.15 (exact)	Tiesinga et al. (2021)	all

D.2 Units and conversions

Quantity	**Conversion**
Pressure	$1\ \mathrm{hPa} = 1\ \mathrm{mb} = 100\ \mathrm{Pa}$; $1\ \mathrm{mmHg\ (torr)} = 1.333\ \mathrm{hPa}$
Temperature	$T_\mathrm{K} = T_{\circ\mathrm{C}} + 273.15$; $T_{\circ\mathrm{C}} = \frac{5}{9}\,(T_{\circ\mathrm{F}} - 32)$
Temperature difference	$1\ \mathrm{K} = 1\ ^\circ\mathrm{C} = 1.8\ ^\circ\mathrm{F}$
Wind speed	$1\ \mathrm{kt} = 0.514\ \mathrm{m\,s^{-1}} = 1.852\ \mathrm{km\,h^{-1}}$
Length	$1\ \mathrm{km} = 1000\ \mathrm{m}$; $1\ \mu\mathrm{m} = 10^{-6}\ \mathrm{m}$
Power per area	$\mathrm{W\,m^{-2}}$: watts per square metre

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